Accounting for Empowerment?
Examining Women's Financial Inclusion in India

Tarana Chauhan*

May 8, 2026

Abstract

Bank accounts are an essential first step towards formal savings and credit in most countries, yet their impact on women's control over resources remains under-explored. I investigate the effects of the 2014 policy in India that provided free bank accounts and led to an unprecedented increase in women's account ownership. This paper shows that women's account opening improved households' financial access but did not improve women's independent decision-making. Using a difference-in-difference estimation that exploits the sharp timing of the policy and a high-frequency household panel data, I find that women's account ownership increased household's likelihood to save in formal instruments and switch to formal sources of borrowing but did not affect consumption patterns consistent with women's preferences. Exploiting regional variation in pre-policy bank infrastructure, I further analyze the effects on women's self-reported decision-making. Districts with faster account expansion did not exhibit overall improvement of women's participation in household purchase decisions or spending autonomy. These results highlight that although bank accounts are a low-cost tool to promote household savings and credit access, they may not directly enhance women's control over resources.

Keywords: Bank Account Ownership; Household Resource Allocation; Women's Decision Making; Government Policy; Women's Empowerment; India

JEL codes: D13; D14; G21; G28; G51; I38; J12; J16; R28

*Postdoctoral Research Associate, Center for Philosophy, Politics, and Economics, Brown University, Providence, Rhode Island 02912, U.S.A.; tarana_chauhan@brown.edu. I would like to thank Christopher B. Barrett, Brian Dillon, Carolina Castilla and Francine Blau for their comments at various stages of this paper. I am also grateful to the conference participants at GLO-JOPE, Indian Statistical Institute, North East Universities Development Consortium and European Winter Meeting of the Econometric Society and seminar participants at Cornell University, Colgate University and Lafayette College.

1 Introduction

In the last decade, policymakers promoted savings and credit through bank accounts as access helped reduce poverty and debt, smooth consumption and increase investment (Burgess and Pande, 2005; Brune et al., 2016; Pomeranz and Kast, 2024). For women, account ownership increased their labor supply (Field et al., 2021), savings and investment (Dupas and Robinson, 2013), and perceived financial well-being (Prina, 2015). Impacts on women's decision making are varied where some *resource intensive programs* had positive impacts through training or social cohesion (Ashraf et al., 2010; Desai and Joshi, 2013; Kumar et al., 2021) but not always (Field et al., 2021). Meanwhile, most account expansion efforts in low- and middle-income countries focus on zero-fees instead of financial literacy and training. Given the rapid increase of this type of no cost bank accounts between 2011 and 2022 (28% to 66% for women, World Bank 2022), a central question is how do such accounts affect household behavior and women's agency.

To answer this, I examine the effects of India's large-scale provision of no-fee bank accounts on household resource allocation and married women's participation in financial decisions. Nationally representative data shows that unbanked, married women had lower decision-making power than banked women in 2011-12: 48% vs. 55% report participation in decisions on purchase of large household items and 87% vs. 96% report autonomous spending (India Human Development Survey, Desai et al. 2018). If account ownership improves women's agency over spending and saving decisions, the policy should impact how households allocate resources and women's reports of participation in decision making.

I examine this by evaluating the 2014 account expansion policy implemented by the Government of India. The policy enabled no-fee bank accounts for the unbanked with minimal proof of identification to increase financial access. At the time of policy implementation, about 40% of adults did not have a bank account, and the majority of them were women (Centre for Monitoring Indian Economy, 2025). Within a week, 18 million new accounts were opened (Government of India, 2014) and, a year later, 69% of women had a bank account - a significant jump from the 45% before policy (Centre for Monitoring Indian Economy, 2025). World Bank's Findex (2022) reported that the gender gap in account ownership dropped from 18 percentage points to 3 between 2014 and 2017. The policy's disproportionately large effect on women makes it well suited to studying the consequences of women's account ownership using quasi-experimental methods.

I use two empirical models, one analyzing changes at the household level and the other at the district. In the household analysis, I use a difference-in-differences (DiD) model that estimates the effects of women's account ownership on household resource allocation. I restrict the sample of a high-frequency longitudinal survey to households where the wife of the household head was unbanked before the policy. Then, I compare households in which the wife opened a bank account as soon as the policy was implemented with households where she remained unbanked for at least one year after. Men's account ownership was universal in this sample, allowing me to isolate the impact of wife's account on the household[1]. I apply inverse probability weights by re-weighting households on the wife's likelihood of being banked using observable characteristics to address potential selection bias between the two groups. Leveraging the sharp timing of the policy, I estimate differences between the two groups before and after policy implementation. The outcome variables include households' take up of saving instruments, formal and informal credit and monthly consumption allocations. Conditional on household characteristics, if wife's account creation shifted resource allocation, this suggests that she previously had limited control over household financial decisions.

The district-level analysis expands the outcome variables to women's self reported decision making and connects this study with others on women's empowerment. I examine whether new bank accounts improved women's decision making using a separate DiD model that leverages a spatial variation in the expansion of bank accounts. The policy led to differential rates of account expansion by ex-ante bank infrastructure. I exploit this differential trend to examine aggregate changes in women's decision-making with a balanced district panel that combines administrative banking data with nationally representative household surveys. The estimation controls for pre-existing differences in variables correlated with bank infrastructure and women's empowerment. To support the identifying variation, I confirm that account expansion in districts with above- versus below-median bank infrastructure was parallel before the policy. There was 10% faster expansion of bank accounts in districts below median bank infrastructure than the rest after the policy was implemented.

My analyses produce three main sets of results. First, women's account ownership increased households' saving and borrowing through formal channels. When the wife of the household head opened a bank account, households were 11.5 percentage points more

[1] The policy governed the opening of bank accounts and served as an exogenous shock to households' *take-up* of bank accounts. Therefore, the analysis estimates the effects of *account ownership*. The policy did not mandate transactions therefore account utilization is not tested directly but examined through heterogeneous analyses.

likely to save in formal instruments such as fixed deposits and government bonds. There was a shift towards formal credit: households were 0.5 percentage points more likely to borrow from formal sources while credit from informal sources reduced by 0.7 percentage points. Therefore, the addition of wife's account helped meet households' demand for institutional saving and credit.

Second, while the likelihood to save increased, account ownership did not shift household consumption in favor of women's preferences. There were no significant changes in household's consumption of goods reflecting women's choices such as clothing, footwear, beauty products, and services.

The third result examines women's self reports of involvement in spending decisions. There were no aggregate improvements in women's participation in decisions around large household purchases in districts with faster account expansion following the policy. There was a relative decline in women's spending autonomy in the faster growing districts due to decline in employment. Heterogeneity analysis suggests that women's pre-existing mobility and employment correspond positively with her decision making. Household's trust in banking corresponds positive with women's participation in large purchase decisions but negatively with her spending autonomy. The heterogeneous effects are large relative to the control group, but statistically insignificant.

Women are more likely than men to have inactive accounts[2] in India (Demirguc-Kunt et al., 2022). Therefore, along with a bank account, improving women's mobility, employment and trust in institutions are relevant for women's agency in spending decisions.

This paper makes three important contributions. First, it advances research on women's financial inclusion and empowerment. Studies have examined the economic and social impacts of microfinance institutions (Chliova et al., 2015; Kochar et al., 2022), bank branches (Garg and Gupta, 2021; Bhukta et al., 2024), and digital financial services (Karlan et al., 2016; Schaner, 2016; Suri, 2017; Toth and Greenland, 2023; Riley and Shonchoy, 2024). This paper estimates the effect of account ownership while holding bank infrastructure constant. It directly builds on the work of Field et al. (2021), who found that bank accounts increased female labor supply but did not improve women's say on how their money is spent in four districts of Central India. In this paper, I investigate why women's involvement in spending decisions is not impacted from account ownership by examining effects on household resource allocation. I also expand the analysis to pan-India using a nationally representative sample.

The broader empirical literature is divided on whether financial interventions improve

[2] Dupas et al. (2014); Karlan et al. (2014); Brune et al. (2016) report that lack of trust in banks and transaction costs such as transportation result in low account use.

women's decision-making. Microcredit interventions have generally found no significant effects on women's decision making in India (Banerjee et al., 2015), rural Mongolia (Attanasio et al., 2015) or Morocco (Crépon et al., 2015); although interventions with a treatment arm explicitly targeting women's empowerment have shown more promise (Angelucci et al., 2015). This paper shows that low-cost government financial inclusion efforts improved household's saving and borrowing behavior but did not meaningfully improve women's independent decision making.

Second, this paper adds to the literature on women's asset ownership and household consumption allocations. Earlier studies document that increases in women's asset ownership are associated with higher spending on non-durable goods, time-saving appliances, health, education, and food (Beegle et al., 2001; Quisumbing and Maluccio, 2003; Duflo and Udry, 2004; Mishra and Sam, 2016). A multi-country analysis finds that women's ownership of land or housing increases participation in decisions on healthcare, household purchases and mobility (Amir-ud Din et al., 2024). In contrast, I find that women's ownership of a bank account corresponds with increased use of formal saving instruments and higher health spending, but no change in consumption of food, clothing, or jewelry, and a decline in education expenditure.

Finally, this paper presents new evidence on the impacts of the largest account expansion policy of the 21st century. Agarwal et al. (2023) analyzed 1.5 million accounts opened under this policy and found low deposit and withdrawal activity in the first six months. The newly banked households improved their ability to manage liquidity in response to shocks such as marriage or rainfall. Similarly, an experiment in Nepal offering comparable accounts to female-headed households found that while first-time bankers had limited saving capacity (Prina, 2015), account ownership influenced consumption patterns, coping ability, and perceptions of financial security. This paper builds on these insights by studying not only household saving behavior but also consumption allocations and self-reported spending decisions.

2 Context

The Reserve Bank of India (RBI)'s first consolidated efforts to increase account ownership began in an advisory capacity. In 2005, it recommended commercial banks to provide a "no-frills account" that required low or no minimum balance. This was later restructured into the "Basic Savings Deposit Account" in 2012 to include digital services like ATMs and electronic payments[3]. Despite these initiatives, only three out of five men and two out of five women (altogether, 47% of adults) reported a bank account at the end of 2013 (Financial Inclusion Insights (FII), Intermedia 2014). Important reasons for low account ownership included costs of maintaining an account, lack of documents and limited trust in institutions[4].

2.1 Account expansion policy

The Indian Central government addressed these challenges to account ownership through a policy implemented in end-August 2014. Under the Pradhan Mantri Jan Dhan Yojana (PMJDY), banks could not charge account opening or maintenance fees to the account holder and beneficiaries were required to provide only one government validated identification. The account could be opened at a bank branch or in the beneficiary's village through banks' field agents called Business Corespondent (BC)[5]. The account included additional services such as a debit card free of charge, an accident insurance of USD 1,638 (in nominal terms, USD 5,439 in purchasing power parity (PPP) terms[6]) and an overdraft facility of up to USD 82 (in nominal terms, 272 in PPP) after six months of satisfactory savings/credit performance. Other components of this policy are described in the Online Appendix.

The first phase of the policy was implemented nationwide from August 2014 to August 2015 excluding states in hilly region facing infrastructure and connectivity constraints[7], and districts affected by armed insurgency[8]. By March 2015 (the end of the 2015 financial year), 147 million accounts were opened under this policy, of which 90% were issued a debit card (Ministry of Finance, 2024). The average account balance was INR 1065 (approximately USD 45.5, purchasing power parity). The PMJDY accounts were 10% of the total savings and commercial bank accounts reported by RBI in March 2015. Appendix figure B.1 shows that the largest increase in all bank accounts between 2007 and 2018 in the Phase 1 regions was in the first year of policy. The PMJDY had a comprehensive scope helping individuals store money, transact using ATM and allowing an overdraft for non-business purposes. While the policy did not target women, it offered the potential for women (both income-earners and unemployed) to improve their control over resources

[3] Section 1 in the Online Appendix describes the policy efforts.
[4] Responses from World Bank's 2011 Findex survey and 2014 FII are summarized in Figures 1 and 2 of the Online Appendix.
[5] Section 1.2 in Online Appendix describes the role of the BC.
[6] Estimated using OECD Data on Purchasing Power Parities available here.
[7] These include states in the North East - Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland and Tripura - as well as parts of Himachal Pradesh, Jammu & Kashmir and Uttarakhand.
[8] A list of 35 districts worst affected by "Left-Wing Extremism" is available here.

through these complementary features. Appendix figure B.2 shows that the gender gap in account ownership reduced significantly after 2014.

The following advancements in financial inclusion are not expected to confound the results of this paper[9]. First, the government's efforts to deposit welfare payments directly into bank accounts starting in 2013 were expected to boost the utilization of bank accounts opened under PMJDY. However, the expansion of digital payments was slow between 2013 and 2016. Only 16% of the targeted schemes were covered by December 2016 (Srinivas and Kapur, 2017). For the three schemes[10] that constituted 90% of digital welfare payments, only 50-60% of the total allocations were transferred digitally. Second, the penetration of ATMs, internet and mobile banking services was limited around the time of PMJDY implementation. The annual transaction values through ATMs and mobile banking were a fraction (25% and 1%, respectively) of the value of total bank deposits in 2014. Third, the monetary policy that de-valued currency and introduced high denomination notes, influencing the greater use of digital platforms in November 2016, has a negligible overlap with the post-policy measurement of women's empowerment in this paper[11].

3 Women's empowerment and conceptual framework

3.1 Defining women's empowerment

Household resource allocation

Drawing on insights from the literature, I investigate changes in household's decision to save, borrow and invest as well as consumption allocations in response to wife's account ownership. Bank accounts typically increase savings, and evidence from Kenya shows that women entrepreneurs were more likely to save than men when provided with a savings account (Dupas and Robinson, 2013) or in response to positive income shocks (Robinson, 2012). I test uptake of formal and informal credit as rising account ownership (77% in 2017) did not improve women's access to bank credit (7%) in India (Chavan, 2020). I also investigate household's investment in two categories. The first is gold and related assets which is preferred by women as an asset accumulation strategy (Antonopoulos and Floro, 2005; Quisumbing, 2011); and the second includes mutual funds, private equity and real estate. This analysis is in line with evidence of gender-segregated investment patterns in

[9] See Section 1.4 of the Online Appendix for descriptive evidence.

[10] Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) and Liquefied Petroleum Gas (LPG) subsidies and pensions under the National Social Assistance Programme (NSAP)

[11] Less than 0.6% of the survey observations on decision making were collected in November and December of 2016.

rural Indonesia (Pangaribowo et al., 2019). More women owned jewelry while more men owned non-agricultural land and house.

Finally, the literature suggests that improvements in women's bargaining power within the household affect consumption allocations. For example, women's asset ownership is associated with increased spending on clothing, time-saving appliances, jewelry, food, health, and education (Beegle et al., 2001; Quisumbing and Maluccio, 2003; Duflo and Udry, 2004; Mishra and Sam, 2016). However, effects may differ by asset type. Pangaribowo et al. (2019) find that women's ownership of liquid assets such as jewelry is positively associated with spending on high-nutrient foods, while ownership of non-liquid assets doesn't have a similar effect. In consumption allocations, I analyze household expenditure on items strongly correlated with i) women's private consumption such as clothing, footwear, accessories, cosmetics and beauty services; ii) women's domestic chores such as cooking and cleaning (expenditure on food, utensils and cleaning products); and iii) consumption that substitutes women's time spent on domestic chores (appliances and domestic labor). I also investigate the consumption of items dominated by male preferences such as tobacco products and liquor, shaving articles and fuel for purposes other than cooking[12]. Lastly, I test for household expenditure on education.

Decision making

I supplement the above analysis using two direct measures of women's involvement in decision making: whether they participate in decisions on large household purchases (individually or jointly with spouse), and whether they have money available to use autonomously. Unlike decision making on food preparation or daily household purchases, these variables are not correlated with the stereotypical division of labor between household members and less likely to be delegated to women when the opportunity cost of time spent on decision making is high (Kochar et al., 2022). Instead, these financial decision making variables are more directly influenced by women's ability to save. The survey data show few reports of solo decision-making on large household purchases. Therefore, I combine women's solo and joint decisions with their spouse on large household purchases into a single variable.

Survey based measures of women's decision making ability and household's gender attitudes are frequently used in empirical works examining women's empowerment (Heckert

[12] Men are more likely to consume tobacco products and alcohol (Cawley and Ruhm, 2011; Oncini and Guetto, 2018) and have a more inelastic demand than women (Nelson, 2014). Fuel consumption captures usage for automobiles and all purposes other than cooking. With the share of licensed female drivers in India less than 7% as recently as 2023 (Ministry of Road Transport and Highways (India), 2023), fuel expenditure is also expected to be driven by men's demand.

et al., 2023; Doss et al., 2022; Kosec et al., 2022; Annan et al., 2021; Field et al., 2021; Swaminathan et al., 2012; Connelly et al., 2010; Allendorf, 2007)[13]. Financial interventions such as loan provision (Kochar et al., 2022) and poverty alleviation and relocation programs (Ding et al., 2024) have improved women's decision making ability on household durables and routine purchases, clothing, food expenditure and borrowing/lending decisions, and investment in children's education. This paper provides insights on larger household expenses and availability of resources to spend autonomously.

3.2 Potential effects of account ownership on women's empowerment

Bank accounts are likely to increase women's propensity to save (Dupas and Robinson, 2013). Increased savings or control over money through account ownership can improve her participation in household's purchase decisions and spending autonomy as well as change consumption allocations. However, in the setting of limited agency and mobility, women may not be able to fully realize these gains. There are three potential channels that can determine how account ownership affects married women's empowerment: spousal relations, account operability and frequency of transactions.

First, cooperation between spouses is essential to women's control over their bank account in patriarchal societies such as India's. Greater cooperation among spouses in decisions may reduce women's autonomy or cooperation may reduce spouse's controlling behaviors. Ashraf (2009) find that communication increases resources that men allocate to their wives. This is an important channel regardless of whether women's accounts are individually or jointly owned.

Second, women's ability to operationalize her account is key to realizing her gains from financial inclusion. This ability is dependent on her mobility outside home as well as financial literacy. Greater autonomy in physical mobility increases her likelihood to visit her bank branch, while financial literacy improves her autonomous handling of the bank account.

Third, a bank account serves as a liquid asset if there is positive account balance and easy account operability. Frequent transactions from the bank account may improve women's autonomy in saving and spending. Both wages and non-labor income can positively impact women's account balance and transactions, therefore, employment and receipt of transfers are important mechanisms. I also investigate household's trust in

[13] In these papers, women are asked about their participation in decisions on food purchase and preparation, control/ use of income, employment, productive decisions in agriculture, mobility, attitudes towards gender equality, borrowing, lending, health, attitudes towards/experience of violence.

banking institutions as Pelras and Renk (2023) showed that trust in institutions affects uptake of services.

4 Data

4.1 Household panel

I analyze household's resource allocation - saving, borrowing and consumption - using the world's largest household panel, Consumer Pyramids Households Survey (CPdx). Starting from January 2014, this survey interviews households every four months. This allows the use of the first two survey waves to analyze pre-policy trends (January - April and May - August 2014). I use the three survey waves after policy enforcement (September - December 2014, January - April 2015 and May - August 2015) to estimate short-term effects of the policy. My sample consists of 13,148 households where the wife of the household head, also the eldest woman, doesn't have a bank account. This sample excludes households that were in the top 5% of monthly expenditure distribution at least once (across the 5 survey waves), relocated to another district, reported no adult member, and where the household head changed during the analysis period. Finally, only the districts where the policy was enforced in the first year are included in the analysis (see 2.1). The Online Appendix describes demographic and financial characteristics of the sample by survey wave. Population-weighted averages of the household panel show that account ownership by men was saturated at the time of policy implementation. The households in the sample had at least 4 members, were predominantly Hindu (about 82%) and resided in towns. The male head of the household and his wife were middle aged at the beginning of the survey, about 42 and 38 years, respectively, and had completed 7 and 5 years of schooling. Almost 70% of the households belonged to backward castes or tribes.

4.2 District panel

The district panel combines cross-sectional household surveys, administrative and census data.

4.2.1 Individual and household surveys

The India Human Development Survey (IHDS) rounds 2004-05 and 2011-12 capture women' self-reports of involvement in decision making before policy while the Demographic Health Survey (DHS) 2015-16 captures post-policy outcomes. Both surveys are nationally representative and their sampling method is described in the Online Appendix. The two outcome variables extracted from IHDS and DHS are - women participating in decisions on large household purchases (alone or jointly), and having money available for autonomous use. I restrict my sample to 328 districts that were enumerated in both surveys and where the policy was implemented in Phase 1[14]. The dataset is further restricted to married women. The Online Appendix summarizes key variables extracted from each of the household surveys. The households in my sample have about 5.5 members, majority have a Hindu household head and reside in rural areas. Respondent women are about 33 years old and have completed 4-6 years of schooling.

4.2.2 Administrative data on bank accounts and branches

I use the district-wise sum of bank branches and accounts to investigate differences in the impact of the policy on account ownership by ex-ante branch density. This data is published at the district level by the RBI in the "Database on the Indian Economy". I use the annual time series for financial years[15] 2006 through 2017 in the analysis. The dataset does not distinguish whether the account is used for savings or business, individual or jointly owned, or the gender of the account owner. It also does not report the volume of transactions. Meanwhile, the total number of bank branches can be disaggregated by their ownership into public, foreign, regional rural, private, and small finance.

4.2.3 Population characteristics

I extract district-level aggregates of total rural and urban population, percent literate, percent scheduled caste/ tribe, number of schools and colleges, access to paved road and electricity from the Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG) dataset. This dataset collates these variables from the 2001 and 2011 Population Censuses.

[14] I match observations to the 2001 Population Census boundaries for consistency between IHDS and DHS. Details in the online Appendix.

[15] A financial year starts April 1 of the preceding calendar year and ends March 31 of the current calendar year. Example, financial year 2006 starts April 1 2005 and ends March 31 2006.

5 Empirical strategy

5.1 Women's account ownership and household resource allocation

In this section, I exploit the sharp timing of the policy to estimate changes in household resource allocation in response to the woman opening a bank account. I use a balanced panel of households where the wife of the household head was unbanked before policy. The sample is also restricted to households where she is the eldest woman in the household to capture effects of the woman with the highest potential to influence household spending[16]. The survey data does not identify bank accounts opened under PMJDY. However, I use the timing of the policy as a shock to preference for a bank account, and analyze the effects of *a new bank account* immediately after the policy was implemented.

The panel observes households eight months (two survey waves) before the policy and twelve months (three survey waves) after. Households where the wife of the male household head opened a bank account in the survey wave after PMJDY was implemented (n=3,302), are characterized as the "treated" group. All households where the wife remained unbanked in the first year of the policy form the "comparison" group[17] (n=9,846). To address selection concerns of household's assignment into treated and comparison groups, I re-weight outcomes based on household's likelihood to be treated. The following equation estimates the likelihood of the wife being banked for all households in the analytical sample using all the demographic variables available in the survey data as well as information on the household's financial access (Table 1). These include religion and caste of household head, household size, sex ratio of adults in the household, urban/ rural location, age and years of completed education of household head and his wife, and the number of women and men in the household owning a bank account, trading account, credit card and mobile phone. The binary variable $D_{hw} = 1$ if the wife opened an account in the survey wave immediately after policy, and 0 otherwise. Matrix $\boldsymbol{X_{hw}}$ is a vector of k covariates. The intercept term is given by β_0 and $\beta_1, ..., \beta_k$ are coefficients of household covariates

$$Pr(D_{hw} = 1|\boldsymbol{X_{hw}}) = \frac{1}{1 + e^{-}(\beta_0 + \beta_1 X_1 + ... + \beta_k X_k)} \tag{1}$$

[16] The literature predicts lower autonomy for women in the household in the presence of mother in-law. For instance, Anukriti et al. (2020) find that mother in-laws limit women's social networks and decision making ability on family planning, and Gram et al. (2018) highlight the negative implications of mother in-laws on women's financial autonomy.

[17] This helps isolate the exogenous timing of the policy within the first year and does not conflate comparison with households where women remain unbanked longer term.

These weights rebalance the comparison group so that it resembles the treated group in terms of observed covariates $\boldsymbol{X_{hw}}$. Comparison households with characteristics similar to treated households receive relatively larger weights, while comparison households with characteristics very different from the treated group receive relatively smaller weights. Table 1 reports pre-policy averages of the covariates in $\boldsymbol{X_{hw}}$ in the original sample (columns 1-5) and after re-weighting using the propensity score weights (columns 6-10). The treated and comparison groups are statistically more similar after this reweighting. Figure 4 in the Online Appendix shows a significant overlap (common support) between the likelihood estimate for households in the treated and comparison groups allowing me to estimate the aggregate treatment effect of wife's account ownership next.

In equation 2, I combine the propensity score weights with a regression adjusted model that estimates differences in the outcome variable pre and post policy ($\triangle Y_{hw} = Y_{hw,post} - Y_{hw,pre}$) for the comparison group units ($C = \mathbb{1}\{D_{hw} = 0\}$) conditional on covariates $\boldsymbol{X_{hw}}$, household (θ_h) and survey wave (ϕ_w) fixed effects. This outcome model captures the counterfactual trend for treated households using comparison group units.

$$m(X_{hw}, \theta_h, \phi_w) = E[\triangle Y_{hw} | \boldsymbol{X_{hw}}, \theta_h, \phi_w, C] \tag{2}$$

The augmented inverse probability weighted DiD model below is doubly robust where if the propensity score is mis-specified (equation 1), the outcome model corrects for confounding (equation 2), and vice versa. $\hat{Y}^{AIPW}$ is the average treatment on the treated group ($D_{hw} = 1$). The estimated probability of treatment $\hat{p}$ (propensity score) is conditional on covariates $\boldsymbol{X_{hw}}$, household (θ_h) and survey wave (ϕ_w) fixed effects.

$$\hat{Y}^{AIPW} = E\left[\frac{\mathbb{1}\{D_{hw} = 1\}}{E(\mathbb{1}\{D_{hw} = 1\})}(\triangle Y_{hw} - m(\boldsymbol{X_{hw}}, \theta_h, \phi_w)) - \frac{\frac{\hat{p}}{1-\hat{p}}C}{E[\frac{\hat{p}}{1-\hat{p}}C]}(\triangle Y_{hw} - m(\boldsymbol{X_{hw}}, \theta_h, \phi_w))\right] \tag{3}$$

Observations in the treated group are weighted by the share of treated households in the sample ($\mathbb{1}\{D_{hw} = 1\}/E(\mathbb{1}\{D_{hw} = 1\})$) for treatment observations reduce the variance in estimation of treatment effects. Observations in the comparison group are re-weighted to represent the treated group in the absence of treatment. These weights are stabilized by the expected odds of treatment for the observations in the comparison group $E[(\hat{p}/1 - \hat{p})C]$. Standard errors are clustered by household to account for serial correlation in outcomes across survey waves within the same household h.

I analyze two types of outcome variables[18]. First, binary indicators of household's sav-

[18] Appendix table A.1 describes each variable in detail.

ing, borrowing and investment behavior. These are - whether household saves in formal sources, saves in microfinance institutes, borrows from formal sources, borrows from informal sources, saves in gold and invests in real estate/ private equity[19]. Second, monthly consumption allocations for women, men and children. I report two scale invariant outcomes[20]: per capital monthly consumption in real terms and share of total monthly consumption[21]. I construct three consumption categories that are likely influenced by women's decision making. The first category includes items most strongly correlated with women's private consumption: clothing and footwear, cosmetics, accessories, and beauty goods and services. The second category includes items that are correlated with women's time spent on domestic chores such as cooking and cleaning. These are food, utensils and toiletries. The third category includes items that help women minimize the time they spend on household chores. These include time saving appliances (e.g. kitchen appliances and washing machine) and services such as domestic help. The fourth consumption category includes items influenced by male preferences. This includes intoxicants such as tobacco products and liquor, shaving articles and fuel for purposes other than cooking. The fifth category is household's total expenditure on education including fees to schools, colleges and private tuition; books; uniforms etc. I report sharpened q values to control for the risk of false discovery rate (Type 1 errors) from testing multiple hypotheses (Benjamini et al., 2006) in each of the results Tables 3 and 4.

Monthly disaggregation of per-capita consumption shows significant pre-trends in the first survey wave between treated and comparison group (results in Online Appendix). This may be the result of noise generated from recall bias as respondents have to report consumption of past four months as the time of interview. To address this, I restrict consumption data of every household to the month prior to interview date. This results in one month's observation for every survey wave. I use equation 3 to analyze effects of wife's account ownership and include month fixed effects instead of survey wave fixed effects to control for seasonal trends in consumption by the month of recall selected.

I include two tests for parallel trends in the outcome variables between the two groups in the pre-treatment time periods. First, I report the chi-squared test for whether treatment effects in all pre-treatment time periods are jointly 0 and the p-value of this estimate in the results (Tables 3 and 4). Second, I report aggregate differences between treated

[19] Using binary dependent variables in equation 3 did not lead to predictions outside [0,1] in the results in Tables C.1 and 3.

[20] Chen and Roth (2023) show that if treatment affects the extensive margin, then rescaling the outcome variable affects the coefficient estimate of the treatment effect.

[21] The per capita specification controls for total monthly per capita consumption and captures level differences conditional on total spending. The share specification captures reallocation across categories.

and comparison groups by survey wave in Appendix Tables C.1 and C.2.

5.2 District-wise account expansion and women's decision making

Identifying the impact of PMJDY on account ownership

Figure 2 shows that the supply of bank branches before the policy was positively correlated with account ownership in a district. Therefore, the provision of no-fee bank accounts is likely to have the largest impact in under-banked districts, viz, districts with lower bank branch availability. I classify districts as having the highest *potential* for impact (henceforth, "High Impact") if their ex-ante branch density[22] (bank branches per 100,000 population) was equal to the state median or below and the remaining as low potential for impact or "Low Impact". Sorting districts by the state median instead of the national accommodates the heterogeneity in levels of bank infrastructure and economic growth between states[23]. The equation below measures annual differences in bank accounts and bank infrastructure between High and Low Impact districts.

$$Y_{dy} = \rho_0 + \sum_{j=2006}^{2017} \rho_j \mathbb{1}(\tau_y = j) \times HighImpact_d + \omega \mathbf{X_{dy}} + \phi_d + \tau_y + \epsilon_{dy} \tag{4}$$

where

$$HighImpact_d = \begin{cases} 1, & \text{if District's bank branch density} \leq \text{State median} \\ 0, & \text{Otherwise} \end{cases}$$

The following outcome variables are analyzed - the total bank branches in a district per 100,000 population, total bank accounts per 100,000 population, and percent change in number of bank accounts since 2014. Differences by the continuous variable of bank branch density are reported in Appendix Table C.3. The data does not identify sex of the primary account owner, therefore, I cannot test for expansion of women's accounts specifically[24]. The coefficient ρ_j estimates mean difference in the outcome variable be-

[22] District-level branch density was reported by RBI in March 2014 and the policy was implemented in August of the same year.

[23] Figure 6 in the Online Appendix summarizes the heterogeneity in the state median values in the sample.

[24] Utilizing the survey data, I find that the proportion of banked women in High Impact districts increased by 34% following the policy (Appendix Table C.4). However, the test for parallel trends in panel A shows significant pre-trends between High and Low Impact districts potentially driving the larger post-policy impact. Additionally, there are discrepancies in survey questions between IHDS and DHS. The IHDS asks whether the woman has her name on a bank account while DHS asks whether she has a bank account *that she uses*.

tween High and Low impact districts for financial years 2006 to 2017. Table 2 describes pre-policy level differences between High and Low Impact districts in population characteristics extracted from the Population Census in 2011. This motivates the inclusion of covariates that predict each district's classification as High/Low Impact from both population variables in 2011 and inter-temporal differences since the 2001 Census. A subset of these covariates are included as $\mathbf{X_{dy}}$ in equation 4 using post-double-selection LASSO method[25]. I include unit (district) fixed effects, ϕ_d, and time (financial year) fixed effects, τ_y, and cluster standard errors at the district level. As the treatment rule is specified by state median bank infrastructure, I also report differences between High and Low Impact districts within state-year by using state $\times$ year fixed effects.

Women's decision making in High versus Low Impact districts

Equation 5 estimates the effect of account expansion on women's empowerment between High and Low Impact districts in response to the policy. Empowerment in this model is measured using two binary variables - women reporting participation in decision making on large household expenditures (alone/ jointly with spouse), and having spending autonomy[26]. Due to survey design, this test is restricted to married women between ages 15 and 49. A difference-in-difference estimation eliminates the residual differences between districts that explain women's empowerment levels. Table 7 tests the identifying assumption that empowerment outcomes between High and Low Impact districts did not vary before policy. Verifying that women's empowerment evolved statistically similarly between the two district types, the coefficient β_1 in the equation below captures the effect of unanticipated acceleration of bank accounts on women's empowerment. These estimates are conditional on differences in the ex-ante characteristics of the High and Low Impact districts summarized in Table 2.

$$Empowerment_{idst} = \beta_0 + \beta_1 (HighImpact \times Post)_{dst} + \boldsymbol{\mu} \boldsymbol{Z_{idst}} + \zeta_t + \eta_d + \pi_{s \times t} + \epsilon_{idst} \tag{5}$$

The variable $HighImpact = 1$ for districts with bank branch density equal to the state median or below at the time of policy implementation, and 0 otherwise. The matrix

[25] The post-double-selection methodology by Belloni et al. (2013) runs two Least Absolute Shrinkage and Selection Operator (LASSO) regressions and reports the intersection of controls that are significantly correlated with the dependent variable in both estimations. The first regression tests for effects of all potential controls on the outcome variable in equation 4 and the second tests the effect of the same controls on the binary variable of High/Low impact. This is operationalized using the STATA module pdslasso.

[26] Although equation 3 estimates the impact of account expansion on binary dependent variables, Tables 7 and 8 do not report predictions outside [0,1].

$\mathbf{Z_{idt}}$ includes women and household-specific covariates selected using post-double selection method (Belloni et al., 2013)[25]. The model controls for district-wise differences in the outcome variables with district fixed effects (η_d) and inter-temporal differences using year fixed effects (ζ_t). Columns 2 and 4 in Tables 7 and 8 also control for state-level confouding variation such as bank infrastructure, governance, economic growth and women's empowerment with state $\times$ year fixed effects ($\pi_{s \times t}$). Finally, standard errors are clustered at the district level which is the level of treatment assignment in this model and error terms are likely correlated within a district. In this analysis, empowerment outcomes and covariates are observed over the calendar year t instead of financial year y.

I test the robustness of this specification using alternate definitions of High/ Low Impact districts in Appendix Table C.3. These include within state presence of public or private sector bank branches, and dis-aggregating results by deciles of bank branch density instead of median.

Other outcomes of women's empowerment

I apply the DiD model in equation 5 to other measures of women's empowerment[27] such as their employment, fertility, mobility outside home, and dowry-related violence by spouse or his relatives[28]. These additional results help verify whether account expansion impacted other non-financial outcomes. Domestic violence is likely under-reported to the police, therefore, the analysis will provide a lower bound estimate of the impact of account ownership on violence against women linked with dowry payments.

Mechanisms

I also explore different factors that may interact with account ownership to impact women's empowerment - control over the account, account operability and balance. There is high prevalence of joint ownership among married couples in India but no public dataset that identifies bank accounts as individual-or-jointly owned. Therefore, I investigate women's cooperation with their spouse and perceptions on wife beating as proxies for women's control over her bank account. As outlined in Section 3.2, greater cooperation with spouse could improve or worsen women's control over her account while acceptability of wife beating will more likely signify less control. Both variables are constructed from indicators using IHDS 2012[29]. Next, I similarly construct two district-level measures of women's account operability. Visiting the bank branch can be a significantly

[27] All variables are extracted from the household surveys IHDS and DHS, except the indicator for dowry-related violence that is extracted from annual reports of the National Crime Records Bureau.

[28] Dowry is the transfer of wealth, property, or goods from the bride's family to the groom or his family at the time of marriage. Such transfers are prohibited in India by law since 1961.

[29] Definition of each variable is provided in Appendix Table A.3.

larger transaction cost for women than men. While the IHDS does not directly measure mobility related to bank access, it captures a number of situations in which women can leave their home without seeking permission from their spouse or other family members that I use to construct a score of her mobility. Another factor that impacts women's ability to operate their account is financial literacy. I use years of completed schooling as a proxy for this. Finally, since account balance and transactions are not directly observable, I infer account use through ex-ante measures of women's employment, household's confidence in banks and household's receipt of government transfers.

6 Results

6.1 Effect of wife's account ownership on household's resource allocation

The addition of the wife's first bank account improved household's engagement with formal financial institutions to save and borrow. Their likelihood to save through bank and post office deposits or government saving bonds increased by 11.5 percentage points (Table 3). The treated group was also 1.6 percentage points more likely to save in gold and related assets, a traditional form of saving considered more directly reflective of women's preferences. These effects grew larger with each post-policy survey wave (Appendix Table C.1). The increased prevalence of savings is consistent with evidence that women are more likely to save when provided commitment savings accounts or following positive income shocks (Ashraf et al., 2010; Robinson, 2012). Since the outcome variable captures household-level rather than individual saving decisions, the estimate should be interpreted as a lower bound on women's preferences for formal saving. Saving in SHGs and other MFIs increased significantly over time, though these estimates should be interpreted with caution given statistically significant pre-policy differences between the treated and comparison groups.

Wife's account ownership also shifted households' borrowing behavior. Uptake of formal loans from banks, registered companies, and employers increased by 0.5 percentage points, while borrowing from informal sources such as moneylenders, shops, and relatives declined by 0.7 percentage points. Investment in real estate and private equity did not satisfy the pre-policy parallel trends test, and there were no significant differences post wife's account opening.

Table 4 reports the effects of wife's account ownership on household consumption expenditure. Households in the treated group allocated a larger but statistically insignificant amount in both per capita and share terms[30] to clothing, footwear, and accessories, a category I attribute to women's preferences (column 1). This suggests no meaningful reallocation toward women's consumption. Estimates by survey wave indicate that the treated group spent significantly more on this category in the survey wave where the wife opened her first bank account but this gap dissipated in subsequent waves (Appendix Table C.2).

The treated group reduced its allocation to consumption associated with men's preferences (column 4, Table 4). The per capita estimate declined by 2.2%, presumably driven by reduced fuel expenditure as this group also reported a larger reduction in per capita transportation expenditure relative to the comparison group (Table 5). While the per capita specification did not satisfy the parallel trends test, the expenditure share specification did. The share of men's consumption expenditure was 3% lower in treated-group households following the policy[30], and this gap persists across multiple post-policy survey waves (Appendix Table C.2).

The literature evaluating non-income transfers to women finds contrasting effects on education. Following account opening, the treated group spent 6% less per capita on education (21% reduction in share of their total monthly expenses). This is consistent with a negative correspondence between women's assets at the time of marriage and the expenditure share on education documented in Indonesia (Quisumbing and Maluccio, 2003). However, contrasting evidence from Bangladesh and South Africa suggests that contextual factors mediate this relationship. In India specifically, women's representation in local leadership positions reduced the gender gap in adolescents' education (Beaman et al., 2012), although education was not a primary focus of public goods provision by elected women representatives (Chattopadhyay and Duflo, 2004).

Household per capita expenditure on items associated with domestic chores, specifically cooking and sanitation supplies, fell significantly among treated-group households relative to the comparison group. Although treated households spent less in absolute terms, their expenditure share was significantly larger than the comparison group before policy, and continued to increase after the policy. As expected, cooking and sanitation

[30] The sharpened q-value for the share estimate approaches conventional thresholds as the procedure controls the false discovery rate (FDR) across the family of hypotheses being tested simultaneously, exploiting the joint distribution of p-values across all outcomes. In this case, strong evidence on other outcomes in the family raises confidence that a hypothesis with a borderline p-value is also a true effect rather than a false discovery.

expenditures contracted proportionally less than total household expenditure. Consumption of time-saving goods and services (column 3) did not satisfy the parallel trends test in either specification and is therefore excluded from this discussion.

Table 5[31] reports consumption patterns across additional expenditure categories. Wife's account ownership shifted spending away from restaurants, transportation, and communication toward healthcare and entertainment. Per capita expenditure on movies, concerts and sports increased by 33% (column 3). Greater healthcare allocation in the treated group (column 7) was consistent across both specifications with a 23% increase in per capita terms.

The combined evidence on saving, borrowing, and consumption reveals how wife's account ownership improved households' access to financial services. Treated households were more likely to save, particularly in gold, an asset associated with women's preferences, and shifted borrowing toward formal sources. The implications of women's decision making on household consumption is ambiguous. In experimental settings, reduced transaction costs of savings accounts increased daily private expenditure in Kenya (Dupas and Robinson, 2013) and educational expenditures in Nepal (Prina, 2015). In this natural experiment, account ownership did not affect household consumption attributable to women's preferences while spending on men's consumption and education declined. I further examine how account ownership can impact women's decision-making in the following section.

6.2 Policy's effect on account ownership and women's empowerment

Policy-induced acceleration on bank account ownership

Figure 3 plots differential trends in bank infrastructure and account ownership between High and Low Impact districts. Prior to the policy, High Impact districts had fewer bank branches and accounts per 100,000 population (top and middle panels). The bottom panel reports percent growth in bank accounts since 2014, the year of policy implementation. Given pre-existing level differences, account ownership grew faster in High Impact districts following the policy. Growth rates were statistically similar between the two district types before 2014 and diverged thereafter[32]. Account expansion in High Impact districts was about 10 percentage points higher relative to Low Impact districts (Table

[31] Estimates by survey was are reported in the Online Appendix.
[32] See Online Appendix for numerical estimates of year-by-year difference.

6). Since account ownership was closer to saturation in Low Impact districts, the policy had limited impact there. I exploit these differential expansion rates to examine changes in women's decision making.

Appendix Table C.3 checks the robustness of results using alternative definitions of High and Low Impact districts. Columns 1 and 2 classify districts by state median branch density of government-owned and private-owned banks, respectively; column 3 uses the continuous measure of total district-level branch density. Panel A shows that the effect of the policy on account expansion is consistent across definitions in columns 1 and 2. Point estimates are larger for the private bank branch density classification, suggesting that the policy had a significantly larger impact in districts with below median private bank supply. Column 3 confirms a negative linear relationship between pre-existing branch density and the acceleration of account ownership following the policy.

Account expansion and women's empowerment

Table 7 confirms the absence of pre-trends in women's decision making between High and Low Impact districts across two pre-policy survey rounds. Table 8 reports that women's participation in decisions on large household purchases was unaffected by the policy (columns 1-2). The coefficient of the interaction term (β_1) is close to zero, and further attenuates when I include state-by-year fixed effects in column 2, absorbing time-varying differences between states.

Women's spending autonomy declined significantly in High Impact districts relative to Low Impact districts following the policy. The estimated effect of -0.038 (column 4) is significant at the 5% level and corresponds to a 6.2% reduction relative to the comparison group mean of 0.615. This result is plausibly attributable to a significant decline in employment in High Impact districts relative to Low Impact districts following the policy (Table 9). The simultaneous contraction in employment directly constrained women's financial autonomy, consistent with how spending autonomy is measured in the post-policy period. The pre-policy IHDS survey asked whether women could use any cash at hand, while the post-policy DHS survey asked whether women could autonomously use *their own* money.

I verify the robustness of results in Appendix Table C.3 where Panel B shows that women's participation in large household purchases is unaffected by varying the definition of High and Low Impact districts. The aggregate decline in spending autonomy documented in the main results is concentrated in districts with lower private bank branch density (column 2, Panel C). Column 3 indicates that women in districts with high branch density experienced gains in spending autonomy following the policy. Appendix Table C.5 decomposes this result by quartile of district-level branch density and shows that the positive correspondence between branch density and spending autonomy is driven exclusively by the uppermost quartile, implying that only districts with the highest pre-existing bank infrastructure exhibit improvements in women's spending autonomy in response to the policy.

Table 9 reports the impact of the policy on other dimensions of women's empowerment[33]. The policy had no differential impact on fertility across district types. Point estimates are negative, close to zero, and statistically insignificant. Women in High Impact districts reported lower mobility outside their home after the policy. The reduction is statistically insignificant (columns 5-6) but not trivial, representing 20% reduction from the control group mean. The relative decline in mobility in High Impact districts also helps explain the lack of improvements in women's decision making. Police reported crime against women by spouse and family members was approximately 12% lower in High Impact districts than the control group mean (columns 7-8) but the point estimates are not statistical significance.

These results are based on approximately 67,000 women across India and compare directly with Field et al. (2021), in which over 5,800 women in a single Indian state received a bank account, financial training, and direct wage deposits. A key distinction between the two studies is that this paper isolates the effect of account ownership alone, while Field et al. (2021) capture the combined effect of ownership and active utilization. Despite this difference, neither study finds aggregate improvements in women's self-reported decision making. This contrasts with Ashraf et al. (2010), who find that access to a commitment savings account increased women's decision-making power by 0.14 standard deviations in the Philippines, suggesting that incentives to save, rather than account ownership per se, may be the operative mechanism. While this study documents an increased likelihood of saving in households where women opened their first bank account, decision-making outcomes were not measured for the same households, precluding a direct test of this mechanism.

Mechanisms

Tables C.6, C.7 and C.8 examine whether policy effects on women's decision making vary by pre-existing district-level characteristics using a triple interaction term (Characteristic

[33] See Online Appendix Table 5 for the test of parallel trends.

$\times$ High Impact $\times$ Post). Table C.6 finds no heterogeneous treatment effects by ex-ante measures of spousal cooperation or norms around domestic violence. The interaction coefficients are negative and statistically insignificant, with small standardized effects relative to the control group mean, indicating that the intervention did not produce meaningfully differentiated impacts along these dimensions.

Table C.7 examines characteristics likely to affect women's account operability, including mobility and education, and yields mixed results. Women's mobility corresponds positively and significantly with participation in large household purchases in High Impact districts (column 1), though this effect loses statistical significance once state-specific time trends are absorbed (column 2). Mobility, however, cannot be ruled out as an important moderating factor as the standardized effect exceeds 0.1 standard deviations of the control group mean and the minimum detectable effects (MDEs)[34] range from 18 to 20% of the comparison group mean, indicating that the model has insufficient statistical power to detect small or moderate heterogeneous effects. Mobility has no correspondence with spending autonomy. The interaction coefficient is negative, close to zero, and statistically insignificant (columns 3 and 4).

Higher aggregate levels of women's education did not correspond with greater involvement in large household purchase decisions in High Impact districts (columns 5-6). Instead, higher education corresponded with lower availability of money for autonomous spending in High relative to Low Impact districts. This negative correspondence warrants cautious interpretation: The standardized effects are -0.045 SD of the control group mean, suggesting that while statistically significant, the effects are small in economic terms.

Table C.8 examines whether the policy effect on women's decision making is moderated by characteristics that proxy for bank account usage: employment status (columns 1-4), trust in banking institutions (columns 5-8), and receipt of government transfers (columns 9-12). Point estimates of the interaction terms are large relative to the control group mean but statistically insignificant across all three dimensions. The MDEs range from 35% to 73% of the control group mean implying that the study is underpowered to detect effects of the magnitude reported in the first row of the table. Women in High Impact districts with higher pre-policy employment rates are more likely to report spending autonomy following the policy than their counterparts in Low Impact districts, with stan-

[34] I compute MDEs at 80% power ($\alpha = 0.05$) and equal to 2.8 times the standard error of the interaction coefficient.

dardized effects exceeding 0.2 SD of the control group mean. Higher household trust in banking institutions corresponds positively with women's involvement in large purchase decisions but negatively with spending autonomy, suggesting that greater household engagement with formal banking can constrain women's financial autonomy. Although the point estimates are statistically null, the standardized effects of the interaction term are large in columns 5-8 and I cannot rule out that the correspondence is economically meaningful. Finally, the receipt of government transfers corresponds positively but statistically insignificantly with decision making in High Impact districts relative to Low Impact.

7 Conclusion

This paper stems from the global emphasis on financial inclusion as a pathway to alleviate poverty and the use of small-scale experiments to demonstrate the impacts of account ownership on women's economic empowerment and agency. In contrast, there is limited investigation of how large-scale policies, that directly increased bank account ownership in low and middle income countries, impacted women's empowerment. This paper seeks to fill that gap by investigating the effects of India's 2014 bank account expansion policy, which mandated free, on-demand accounts to unbanked individuals. I examine the impact of this exogenous shock to account ownership on household resource allocation and women's decision-making ability.

The paper exploits a high frequency household longitudinal survey and examines changes in household resource allocation when the wife of the household head opened a bank account. I compare households where the wife opened a bank account within four months of the policy's implementation (*treated*) with households where the wife remained unbanked at least a year (*comparison*) after using an augmented inverse probability-weighted DiD. In a separate analysis at the district level, I exploit the pre-policy differences in bank branch density that led to differential trends in growth of bank accounts, and analyze whether unexpectedly higher rates of account ownership impacted women's decision making.

Households with newly banked wives showed an increased propensity to save in formal instruments and gold as well as switch out borrowing from informal sources to formal financial institutions. However, there were no significant changes in household consumption attributable to women's preferences. The district-level analysis demonstrates that the policy accelerated bank account ownership disproportionately in districts where pre-policy branch density was lower and accounts were further from saturation. Comparing districts where the policy had a *higher impact* versus *lower impact*, I find no improvements women's involvement in large household purchase decisions. There was a significant decline in spending autonomy, driven in part by a simultaneous reduction in employment in High Impact districts. The results are robust to different definitions of High and Low Impact. Heterogeneity analyses finds no evidence that the policy effect on women's decision making was moderated by pre-existing spousal cooperation or norms around domestic violence. Women's mobility, employment and household's trust in banking yielded large but statistically imprecise estimates, precluding definitive conclusions about the mechanisms through which account ownership may affect women's empowerment.

This paper's findings are relevant to global policy debates on financial inclusion as the analysis adds empirical evidence from the largest account expansion policy of the 21st century. While policy attention has shifted towards digital financial services to reduce transaction costs and expand access (Lauer and Lyman, 2015; Pazarbasioglu et al., 2020; Brown et al., 2024), traditional accounts remain a key entry point for financial inclusion in many low- and middle-income countries. They are particularly crucial for rural women and low-income households who continue to face barriers to digital adoption. The results of this paper suggest that more targeted interventions are needed to improve women's financial autonomy.

References

Agarwal, S., Alok, S., Ghosh, P., Ghosh, S., Piskorski, T., and Seru, A. (2023). Banking the Unbanked: What Do 280 Million New Bank Accounts Reveal about Financial Access? *Working Paper*.

Allendorf, K. (2007). Do Women's Land Rights Promote Empowerment and Child Health in Nepal? *World Development*, 35(11):1975–1988.

Amir-ud Din, R., Naz, L., and Ali, H. (2024). Relationship between asset ownership and women's empowerment? Evidence from DHS data from 18 developing countries. *Journal of Demographic Economics*, 90(2):154–175.

Angelucci, M., Karlan, D., and Zinman, J. (2015). Microcredit Impacts: Evidence from a Randomized Microcredit Program Placement Experiment by Compartamos Banco. *American Economic Journal: Applied Economics*, 7(1):151–82.

Annan, J., Donald, A., Goldstein, M., Martinez, P. G., and Koolwal, G. (2021). Taking power: Women's empowerment and household Well-being in Sub-Saharan Africa. *World Development*, 140:105292.

Antonopoulos, R. and Floro, M. S. (2005). Asset Ownership along Gender Lines: Evidence from Thailand. Economics working paper archive, Levy Economics Institute.

Anukriti, S., Herrera-Almanza, C., Pathak, P. K., and Karra, M. (2020). Curse of the Mummy-ji: The Influence of Mothers-in-Law on Women in India. *Americal Journal of Agricultural Economics*, 102:1328–1351.

Ashraf, N. (2009). Spousal control and intra-household decision making: An experimental study in the Philippines. *American Economic Review*, 99(4).

Ashraf, N., Karlan, D., and Yin, W. (2010). Female empowerment: Impact of a commitment savings product in the Philippines. *World Development*, 38(3):333–344.

Attanasio, O., Augsburg, B., De Haas, R., Fitzsimons, E., and Harmgart, H. (2015). The Impacts of Microfinance: Evidence from Joint-Liability Lending in Mongolia. *American Economic Journal: Applied Economics*, 7(1):90–122.

Banerjee, A., Duflo, E., Glennerster, R., and Kinnan, C. (2015). The Miracle of Microfinance? Evidence from a Randomized Evaluation. *American Economic Journal: Applied Economics*, 7(1):22–53.

Beaman, L., Duflo, E., Pande, R., and Topalova, P. (2012). Female Leadership Raises Aspirations and Educational Attainment for Girls: A Policy Experiment in India. *Science*, 335(6068):582–586.

Beegle, K., Frankenberg, E., and Thomas, D. (2001). Bargaining Power within Couples and Use of Prenatal and Delivery Care in Indonesia. *Studies in Family Planning*, 32(2):130–146.

Belloni, A., Chernozhukov, V., and Hansen, C. (2013). Inference on Treatment Effects after Selection among High-Dimensional Controls. *The Review of Economic Studies*, 81(2):608–650.

Benjamini, Y., Krieger, A. M., and Yekutieli, D. (2006). Adaptive linear step-up procedures that control the false discovery rate. *Biometrika*, 93:491–507.

Bhukta, R., Pakrashi, D., Saha, S., and Sedai, A. (2024). Does Financial Inclusion Mitigate Social Exclusion? *Working Paper*.

Brown, F., Dimova, M., Kelly, S., and Zhang, X. (2024). How policymakers can leverage digital financial capability for inclusion. Technical report, Women's World Banking.

Brune, L., Giné, X., Goldberg, J., and Yang, D. (2016). Facilitating Savings for Agriculture: Field Experimental Evidence from Malawi. *Economic Development and Cultural Change*, 64(2):187–220.

Burgess, R. and Pande, R. (2005). Do Rural Banks Matter? Evidence from the Indian Social Banking Experiment. *American Economic Review*, 95(3):780–795.

Cawley, J. and Ruhm, C. J. (2011). The Economics of Risky Health Behaviors. In Pauly, M. V., Mcguire, T. G., and Barros, P. P., editors, *Handbook of Health Economics*, volume 2, pages 95–199. Elsevier.

Centre for Monitoring Indian Economy (2025). Consumer pyramids data (cpdx). Accessed: 2025-02-06.

Chattopadhyay, R. and Duflo, E. (2004). Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*, 72(5):1409–1443.

Chavan, P. (2020). Women's Access to Banking in India: Policy Context, Trends, and Predictors. *Review of Agricultural Studies*, 10(1):7–36.

Chen, J. and Roth, J. (2023). Logs with Zeros? Some Problems and Solutions. *The Quarterly Journal of Economics*, 139(2):891–936.

Chliova, M., Brinckmann, J., and Rosenbusch, N. (2015). Is microcredit a blessing for the poor? A meta-analysis examining development outcomes and contextual considerations. *Journal of Business Venturing*, 30(3):467–487.

Connelly, R., Roberts, K., and Zheng, Z. (2010). The Impact of Circular Migration on the Position of Married Women in Rural China. *Feminist Economics*, 16(1):3–41.

Crépon, B., Devoto, F., Duflo, E., and Parienté, W. (2015). Estimating the Impact of Microcredit on Those Who Take It Up: Evidence from a Randomized Experiment in Morocco. *American Economic Journal: Applied Economics*, 7(1):123–50.

Demirguc-Kunt, A., Klapper, L., Singer, D., and Ansar, S. (2022). The Global Findex Database 2021 : Financial Inclusion, Digital Payments, and Resilience in the Age of COVID-19 - Chapter 2 : Use of Financial Services. Technical report, World Bank Group.

Desai, R. M. and Joshi, S. (2013). Collective Action and Community Development: Evidence from Self-Help Groups in Rural India. *The World Bank Economic Review*, 28(3):492–524.

Desai, S., Vanneman, R., and National Council of Applied Economic Research (2018). India Human Development Survey-II (IHDS-II), 2011-12.

Ding, Y., Wang, X., de Brauw, A., and Qiu, H. (2024). Catch up with my husband as I can: Women's decision-making power consequences of China's poverty alleviation relocation program. *World Development*, 173:106433.

Doss, C. R., Meinzen-Dick, R., Pereira, A., and Pradhand, R. (2022). Women's empowerment, extended families and male migration in Nepal: Insights from mixed methods analysis. *Journal of Rural Studies*, 90:13–25.

Duflo, E. and Udry, C. (2004). Intrahousehold Resource Allocation in Cote D'Ivoire: Social Norms, Separate Accounts and Consumption Choices. *National Bureau of Economic Research Working Papers*, 10498.

Dupas, P., Green, S., Keats, A., and Robinson, J. (2014). Challenges in banking the rural poor: Evidence from Kenya's Western Province. *NBER Working Paper 17851*.

Dupas, P. and Robinson, J. (2013). Savings Constraints and Microenterprise Development: Evidence from a Field Experiment in Kenya. *American Economic Journal: Applied Economics*, 5(1):163–92.

Field, E., Pande, R., Rigol, N., Schaner, S., and Moore, C. T. (2021). On Her Account: How Strengthening Women's Financial Control Impacts Labor Supply and Gender Norms. *American Economic Review*, 111(7):2342–75.

Garg, S. and Gupta, S. (2021). Financial Access and Gender Gap in Entrepreneurship and Employment: Evidence from Rural India. *Working Paper*.

Government of India (2014). Guinness World Records Recognises the Achievements made under PMJDY. Technical report, Department of Financial Services, Ministry of Finance.

Gram, L., Skordis-Worrall, J., Mannell, J., Manandhar, D. S., Saville, N., and Morrison, J. (2018). Revisiting the patriarchal bargain: The intergenerational power dynamics of household money management in rural Nepal. *World Development*, 112:193–204.

Heckert, J., Martinez, E. M., Sanou, A., Pedehombga, A., Ganaba, R., and Gelli, A. (2023). Can a gender-sensitive integrated poultry value chain and nutrition intervention increase women's empowerment among the rural poor in Burkina Faso? *Journal of Rural Studies*, 100:103026.

Intermedia (2014). Financial Services Use and Emerging Digital Pathways. Technical report, Financial Inclusion Insights.

Karlan, D., Kendall, J., Mann, R., Pande, R., Suri, T., and Zinman, J. (2016). Research and Impacts of Digital Financial Services. *National Bureau of Economic Research Working Paper*, 22633.

Karlan, D., Ratan, A. L., and Zinman, J. (2014). Savings by and for the Poor: A Research Review and Agenda. *The Review of Income and Wealth*, 60:36–78. Special Issue:Poverty, Development, and Behavioral Economics (Comprises Findings of UNU-WIDER Project New Approaches to Measuring Poverty and Vulnerability).

Kochar, A., Nagabhushana, C., Sarkar, R., Shah, R., and Singh, G. (2022). Financial access and women's role in household decisions: Empirical evidence from India's National Rural Livelihoods project. *Journal of Development Economics*, 155:102821.

Kosec, K., Akramov, K., Mirkasimov, B., Song, J., and Zhao, H. (2022). Aspirations and women's empowerment: Evidence from Kyrgyzstan. *Economics of Transition and Institutional Change*, 30(1):101–134.

Kumar, N., Raghunathan, K., Arrieta, A., Jilani, A., and Pandey, S. (2021). The power of the collective empowers women: Evidence from self-help groups in India. *World Development*, 146:105579.

Lauer, K. and Lyman, T. (2015). Digital Financial Inclusion: Implications for Customers, Regulators, Supervisors, and Standard-Setting Bodies. Technical report, CGAP.

Ministry of Finance (2024). Pradhan Mantri Jan Dhan Yojana (PMJDY) — National Mission for Financial Inclusion — completes a decade of successful implementation. Technical report, Press Information Bureau.

Ministry of Road Transport and Highways (India) (2023). Number of driving licenses issued among females in India in financial year 2020, by state (in 1,000s). Technical report, Statistica.

Mishra, K. and Sam, A. G. (2016). Does Women's Land Ownership Promote Their Empowerment? Empirical Evidence from Nepal. *World Develoment*, 78:360–371.

Nelson, J. P. (2014). Gender differences in alcohol demand: A systematic review of the role of prices and taxes. *Health Economics*, 23:1260–1280.

Oncini, F. and Guetto, R. (2018). Cultural capital and gender differences in health behaviours: a study on eating, smoking and drinking patterns. *Health Sociology Review*, 27(1):15–30.

Pangaribowo, E. H., Tsegai, D., and Sukamdi (2019). Women's bargaining power and household expenditure in Indonesia: The role of gender-differentiated assets and social capital. *GeoJournal*, 84(4):939–960.

Pazarbasioglu, C., Mora, A. G., Uttamchandani, M., Natarajan, H., Erik, F., and Saal, M. (2020). Digital financial services. Technical report, World Bank Group.

Pelras, C. and Renk, A. (2023). When sterilizations lower immunizations: The Emergency experience in India (1975-77). *World Development*, 170:106321.

Pomeranz, D. and Kast, F. (2024). Savings Accounts to Borrow Less: Experimental Evidence from Chile. *Journal of Human Resources*, 59(1):70–108.

Prina, S. (2015). Banking the poor via savings accounts: Evidence from a field experiment. *Journal of Development Economics*, 115:16–31.

Quisumbing, A. and Maluccio, J. (2003). Resources at Marriage and Intrahousehold Allocation: Evidence from Bangladesh, Ethiopia, Indonesia, and South Africa. *Oxford Bulletin of Economics and Statistics*, 65:283–327.

Quisumbing, A. R. (2011). Do men and women accumulate assets in different ways?: Evidence from rural Bangladesh. *IFPRI discussion papers.*

Riley, E. and Shonchoy, A. S. (2024). Encouraging Digital Financial Technology Adoption during a Crisis: Experimental Evidence from Ghana. *Economic Development and Cultural Change (Forthcoming).*

Robinson, J. (2012). Limited Insurance within the Household: Evidence from a Field Experiment in Kenya. *American Economic Journal: Applied Economics*, 4(4):140–64.

Schaner, S. (2016). The Cost of Convenience? Transaction Costs, Bargaining Power, and Savings Account Use in Kenya. *Journal of Human Resources*, 52(4):919–945.

Srinivas, V. and Kapur, A. (2017). Direct Benefit Transfer (DBT), Jan Dhan, Aadhaar and Mobile (JAM). Technical report, Accountability Initiative, Centre for Policy Research.

Suri, T. (2017). Mobile Money. *Anuual Review of Economics*, 9:497–520.

Swaminathan, H., Lahoti, R., and Suchita, J. (2012). Women's Property, Mobility, and Decisionmaking: Evidence from Rural Karnataka, India. *IFPRI Discussion Paper*, 1188.

Toth, R. and Greenland, W. (2023). The Impact of Fast Payment Systems on Financial Inclusion in Emerging Markets: Evidence from India. *Working paper.*

World Bank, . (2022). *The Little Data Book on Financial Inclusion 2022.* World Bank, Washington, DC.

Figures

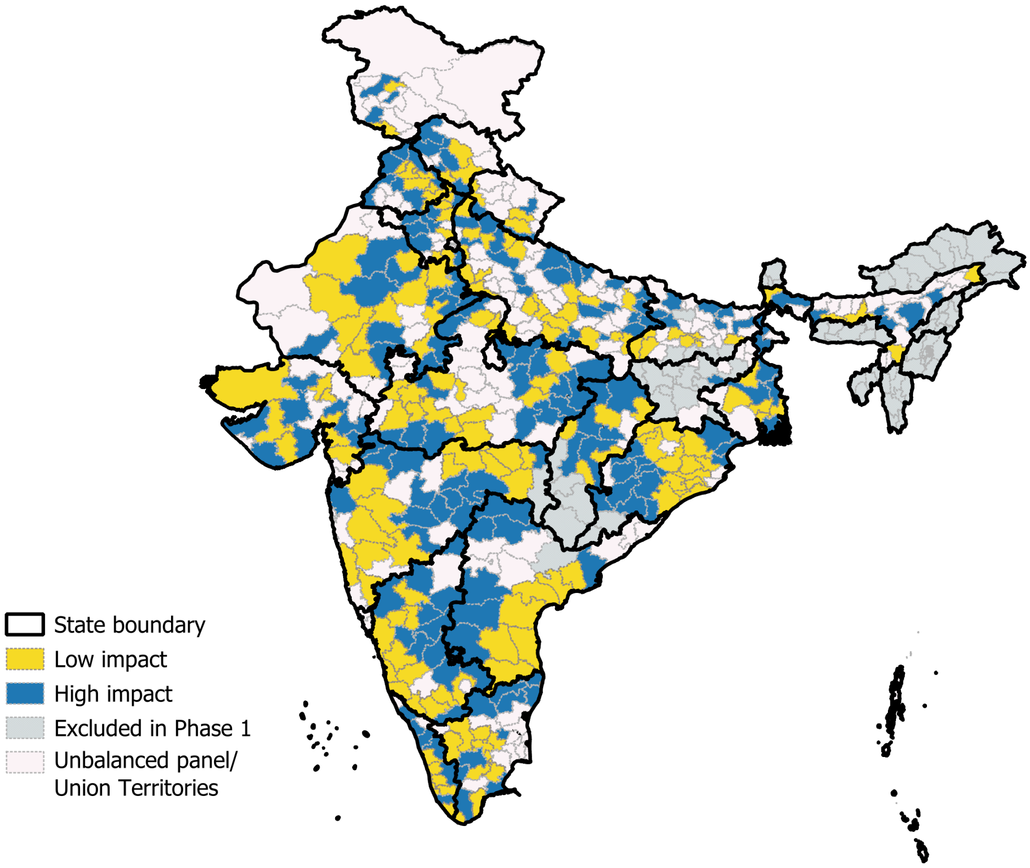

Figure 1: Districts included in sample

Notes: This map of India uses boundaries consistent with the 2001 Population Census. Districts are characterized by their exposure to the policy. "High impact districts" are districts with less than/equal to state median bank branch density in March 2014 and "Low impact" are districts above state median branch density. Districts that were excluded from phase 1 of the account expansion policy are highlighted in gray. Remaining districts in white were excluded from the analysis if they were either not enumerated both before and after policy in the household surveys on women's decision making (unbalanced panel), or part of Union Territories.

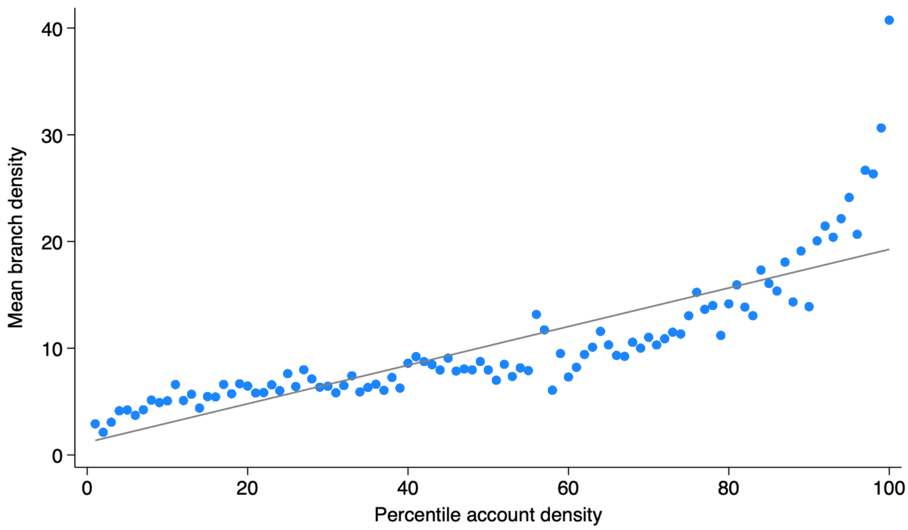

Figure 2: Correlation between bank branch and account density before policy

Notes: The figure plots the correspondence between district-level bank branches and accounts before policy implementation. Districts are binned into percentile scores of total density. Plotting the average bank branches or branch density for each percentile shows that districts with more physical branches had more total accounts per person. The density variables are estimated per 100,000 population. All variables are district-wise estimates from April 2013 until March 2014. Data sources: Reserve Bank of India and Population Census of India 2011. Author's calculations.

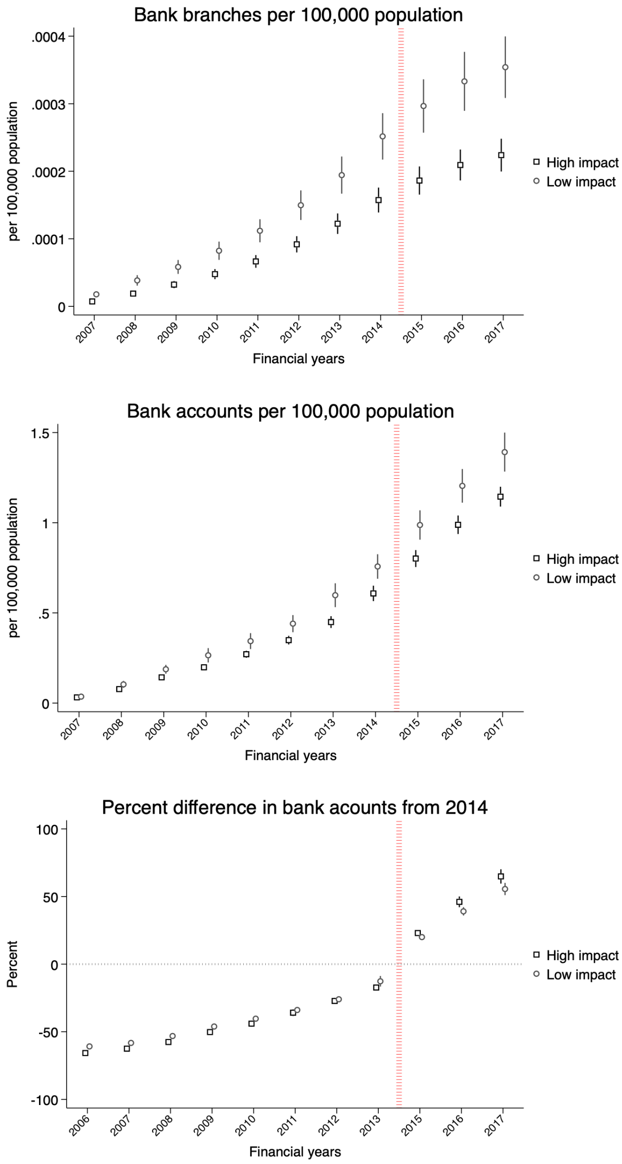

Figure 3: Trends in bank infrastructure and account ownership for High/Low Impact districts

Notes: This figure reports estimated annual trends in bank branch and accounts for High and Low Impact districts. Using Eq. 4, these results control for time-invariant characteristics of a district and variables correlated with both the outcome variable and assignment to High/Low Impact. The top panel plots the estimated bank branch density (total bank branches per 100,000 population) in a district, middle panel reports bank account density, and bottom panel depicts percent difference in accounts since 2014. The x axis depicts financial years where the financial year 2006 includes observations from April 1, 2005 until March 31, 2006, and so on. Each square is the estimated mean of the dependent variable for a High Impact district in that year, while every circle is the estimated mean for a Low Impact district. The vertical solid lines depict 95% confidence intervals and the vertical, dashed line indicates time of policy implementation (August 2014). In the top panels, the financial year 2006 is omitted as reference group while in the bottom panel, 2014 is omitted as the reference year.

Tables

Table 1: Balance of covariates in original and inverse probability weighted (IPW) sample

	Original Sample					Inverse Probability Weighted Sample				
	Treatment		Comparison		t-test of	Treatment		Comparison		t-test of
Covariate	Mean	SE	Mean	SE	difference	Mean	SE	Mean	SE	difference
	(1)	(2)	(3)	(4)	(1)-(3)	(6)	(7)	(8)	(9)	(6)-(8)
Number Children	1.57	(-0.04)	1.67	(-0.03)	-0.1	1.66	(-0.05)	1.64	(-0.04)	0.01
Number daughters	0.73	(-0.02)	0.75	(-0.02)	-0.02	0.76	(-0.02)	0.75	(-0.02)	0
Adult sex ratio	0.98	(-0.01)	0.96	(0)	0.02**	0.96	(-0.01)	0.96	(-0.01)	-0.01
Household size	4.24	(-0.05)	4.38	(-0.04)	-0.14	4.37	(-0.06)	4.33	(-0.04)	0.04**
Number Adults	2.67	(-0.03)	2.71	(-0.03)	-0.04	2.72	(-0.03)	2.69	(-0.02)	0.02
Buddhist (%)	0.85	(-0.4)	0.09	(-0.05)	0.76	0.28	(-0.14)	0.31	(-0.19)	-0.04
Christian (%)	0.55	(-0.16)	0.35	(-0.11)	0.2	0.39	(-0.12)	0.39	(-0.13)	0
Hindu (%)	84.52	(-1.69)	82.47	(-1.66)	2.05*	84.05	(-1.78)	84.02	(-1.52)	0.02
Jain (%)	0.15	(-0.1)	0.04	(-0.02)	0.11	0.07	(-0.04)	0.07	(-0.04)	0
Muslim (%)	10.39	(-1.19)	13.62	(-1.33)	-3.23	11.42	(-1.31)	11.75	(-1.12)	-0.34
Intermediate Caste (%)	8.9	(-1.08)	5.79	(-0.92)	3.11	6.12	(-0.77)	6.99	(-1.09)	-0.88**
Other Backward Caste (%)	40.21	(-1.68)	41.84	(-1.71)	-1.63	41.3	(-1.88)	41.43	(-1.68)	-0.13
Scheduled Caste (%)	23.9	(-1.37)	25.17	(-1.52)	-1.27***	24.9	(-1.51)	24.58	(-1.48)	0.33
Scheduled Tribe (%)	4.04	(-0.68)	4.57	(-0.86)	-0.54**	4.81	(-0.83)	4.37	(-0.81)	0.43
Male head: Education	7.36	(-0.2)	6.13	(-0.23)	1.24***	6.47	(-0.2)	6.5	(-0.24)	-0.03***
Wife: Education	5.17	(-0.21)	3.59	(-0.17)	1.57***	4.09	(-0.2)	4.05	(-0.19)	0.04**
Wife: Age	39.78	(-0.28)	39.52	(-0.29)	0.27	39.71	(-0.3)	39.56	(-0.29)	0.15
Male head: Age	45.23	(-0.34)	44.47	(-0.32)	0.77	44.8	(-0.35)	44.65	(-0.32)	0.15
Urban (%)	73.8	(-2.58)	62.14	(-2.93)	11.67***	64.06	(-3.37)	65.58	(-2.77)	-1.52

(Contd.)

	Original Sample					Inverse Probability Weighted Sample				
	Treatment		Comparison		t-test of	Treatment		Comparison		t-test of
Covariate	Mean	SE	Mean	SE	difference	Mean	SE	Mean	SE	difference
	(1)	(2)	(3)	(4)	(1)-(3)	(6)	(7)	(8)	(9)	(6)-(8)
Quantile of asset index	2.94	(-0.07)	2.43	(-0.07)	0.50***	2.75	(-0.07)	2.5	(-0.07)	0.26***
Male head owns bank account (%)	100	(0)	100	(0)		100	(0)	100	(0)	
Wife has Bank account	0	(0)	0	(0)		0	(0)	0	(0)	
Wife has Credit card	0.06	(-0.04)	0	(0)	0.06	0	(0)	0	(0)	
Wife has mobile phone	23.77	(-2.05)	9.63	(-1.21)	14.15***	19.19	(-1.78)	10.53	(-1.28)	8.65***
Number of household members with...										
Women have Bank account (num)	0.09	(-0.01)	0.04	(0)	0.05***	0.07	(-0.01)	0.05	(0)	0.03*
Women have Credit card (num)	0	(0)	0	(0)	0	0	(0)	0	(0)	
Women have mobile phone (num)	0.35	(-0.03)	0.15	(-0.02)	0.20***	0.28	(-0.02)	0.16	(-0.02)	0.12***
Men have Bank account (num)	1.29	(-0.02)	1.28	(-0.01)	0.01	1.29	(-0.02)	1.28	(-0.01)	0.01
Men have Credit card (num)	0.01	(0)	0.01	(0)	0.01**	0.01	(0)	0.01	(0)	0
Men have mobile phone (num)	1.34	(-0.02)	1.27	(-0.02)	0.08	1.29	(-0.02)	1.28	(-0.02)	0.01**
N	3302		9846			3302		9846		

Notes: The table reports mean values and standard errors of variables in wave 2 (pre-policy) for the (original) sample of households drawn from CPdx (columns 1-4), and the differences between mean values of the treated and comparison group (column 5). Columns 6-9 report means and standard errors of the sample transformed using inverse probability weighting, and column 10 estimates the mean difference. The treated groups includes where the wife was unbanked before policy and opened a bank account in the survey wave after policy (September through December 2014). The comparison group includes households where the wife did not own a bank account before and in the first year of the policy (until August 2015). The sample includes households where the wife of the male household head did not have a bank account in wave 2. In these households, there is no older female present. The following households are excluded from analysis - Households with no adult member, where the head of the household changed or household relocated outside district between waves 1 and 5, members switched out of account ownership and whose total monthly expenditure was above the 95th percentile at least once. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 2: Summary statistics of High/ Low impact districts before policy

Variable	High impact (n=168) Mean	SE	Low impact (n=160) Mean	SE	Pairwise t-test Difference
	(1)	(2)	(3)	(4)	(5)
		Population variables			
Rural (%)	78.03	-1.11	67.09	-1.49	10.94***
Scheduled Caste/ Tribe (%)	29.24	-1.19	24.66	-0.93	4.58***
Literate (%)	60.58	-0.74	66.86	-0.7	-6.28***
Electricity (hours)	3.78	-0.23	3.03	-0.13	0.75***
Paved road (kms.)	3.88	-0.2	3.08	-0.11	0.80***
Number of colleges	322.71	-29.22	313.75	-23.03	8.96
Number of high schools	971.64	-73.19	843.23	-52.98	128.42
Number of middle schools	7225.36	-571.76	5588.43	-313.46	1636.93**
Number of primary schools	11037.29	-908.84	8019.15	-454.32	3018.14***
		Household variables			
Respondent has bank account	0.35	-0.01	0.4	-0.01	-0.05***
Alone/Joint decision on big purchases	0.5	-0.01	0.5	-0.01	0
Spending autonomy	0.9	-0.01	0.91	-0.01	-0.01
Household size	5.62	-0.06	5.44	-0.06	0.18**
Household asset quintile	2.82	-0.07	3.26	-0.06	-0.44***
Female headed household	0.12	-0.01	0.13	-0.01	-0.01
Household head: Schooling	5.4	-0.13	6.19	-0.13	-0.79***
Household head: Hindu	0.83	-0.02	0.83	-0.02	-0.01
Household head: Muslim	0.13	-0.01	0.11	-0.01	0.02
Household head: Christian	0.01	0	0.03	-0.01	-0.02**
Household head: Scheduled Caste	0.22	-0.01	0.21	-0.01	0.01
Household head: Scheduled Tribe	0.09	-0.01	0.07	-0.01	0.03
Head: Other Backward Caste	0.45	-0.02	0.44	-0.02	0.02
Household head: Intermediate Caste	0.19	-0.01	0.24	-0.02	-0.04*
Household head: Upper Caste	0.05	0	0.06	-0.01	-0.01*
Respondent: Age	33.88	-0.12	34.77	-0.13	-0.89***
Age gap between spouses	4.89	-0.11	5.28	-0.12	-0.39**
Respondent: Years of schooling	4.73	-0.17	5.95	-0.18	-1.21
Respondent is employed	0.59	-0.02	0.53	-0.02	0.06**
Respondent: employed in agriculture	0.4	-0.02	0.33	-0.02	0.07**
Respondent has children	0.92	0	0.92	0	-0.01
Number of children	2.54	-0.04	2.39	-0.03	0.14***
Spouse: Years of schooling	6.86	-0.14	7.66	-0.13	-0.80***
Spouse is employed	0.95	0	0.94	0	0.01**
Spouse employed in agriculture	0.31	-0.02	0.26	-0.02	0.06**

Notes: The table reports averages and standard errors of household and population-level characteristics of High (columns 1-2) and Low Impact districts (columns 3-4) before policy. Column 5 reports mean differences between High and Low Impact districts. Household variables are extracted from IHDS 2011-12 while information on population and infrastructure is merged from the 2011 Population Census. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 3: Aggregate treatment effect: Wife's account ownership and household borrowing, saving and investment

	Formal saving	Gold	Formal borrowing	Informal borrowing	Saving SHG/ MFI	Invest-ment
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Treatment	0.115***	0.016***	0.005**	-0.007**	0.008***	0.002
	(0.008)	(0.006)	(0.002)	(0.003)	(0.003)	(0.006)
	[0]	[0.004]	[0.047]	[0.045]	[0.002]	[0.731]
Sharpened q-value	0.001	0.006	0.03	0.03	0.005	0.139
Comparison group mean	0.063	0.055	0.006	0.03	0.003	0.074
Observations	64559	64559	64559	64559	64559	64559
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Joint significance of differences (pre-policy)						
Chi-squared value	0.358	0.001	0.885	0.01	3.131	3.195
p-value	0.549	0.978	0.347	0.922	0.077	0.074

Notes: This table reports the effect of wife's account ownership on household's financial behavior using an augmented inverse probability weighted DiD. The first row reports aggregate treatment effect. The treated group includes households where female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation. The post-treatment dummy variable is assigned 1 for CPdx's surveys 3-5 (September 2014 to August 2015) and 0 for survey waves 1 and 2 (January to August 2014). The dependent variables are specified in column titles and are binary indicators of whether- household has saved in bank/ post office deposits or government bonds (column 1), saved in gold/ related assets (column 2), outstanding borrowing from a bank/ registered company (column 3), loans from friends, moneylender and other informal sources (Column 4), savings in Self-Help group, Chit fund, microfinance institute (column 5) and invested in shares, mutual funds/ real estate (column 6). The second row includes standard errors in parentheses that are clustered by household. The third row reports p values in square brackets and fourth row reports sharpened two- stage q-values that correct the p-value of the interaction coefficient for false discovery rate from testing multiple hypotheses. The last two rows are a test for joint significant of differences between treated and comparison groups before policy reporting the chi-squared statistic and corresponding p value. The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave.The estimation controls for time varying characteristics of households and total monthly consumption expenditure. It also includes household and survey wave fixed effects. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 4: Aggregate treatment effect: Wife's account ownership and household consumption allocations

	Women's consumption (1)	Cooking & sanitation (2)	Time saving (3)	Men's consumption (4)	Education (5)
Panel A: Per capita monthly expenditure in real terms					
Post × Treatment	1.708	-13.044**	-1.119***	-11.657***	-11.315***
	(4.075)	(5.636)	(0.176)	(2.935)	(2.121)
	[0.675]	[0.021]	[0.000]	[0.000]	[0.000]
Sharpened q-value	0.167	0.01	0.001	0.001	0.001
CG mean	397.56	3656.14	2.69	517.7	185.91
Joint significance of differences (pre-policy)					
Chi-squared value	0.009	0.151	11.641	13.544	0.109
p-value	0.923	0.697	0.001	0	0.742
Panel B: Share of total monthly expenditure					
Post × Treatment	0.003	0.015***	-0.0004***	-0.002	-0.006***
	(0.002)	(0.003)	(0)	(0.001)	(0.001)
	[0.112]	[0]	[0]	[0.164]	[0]
Sharpened q-value	0.06	0.001	0.001	0.071	0.001
CG mean	0.058	0.623	0	0.073	0.028
Joint significance of differences (pre-policy)					
Chi-squared value	0.011	3.813	9.241	0.633	2.155
p-value	0.915	0.051	0.002	0.426	0.142
Observations	48165	48165	48165	48165	48165
Covariates	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the effect of wife's account ownership on household's consumption using an augmented inverse probability weighted DiD. Dependent variable is per capital monthly consumption in panel A and share of total consumption in panel B and specified in column titles. The first row reports aggregate treatment effect. The treated group includes households where female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation. The post-treatment dummy variable is assigned 1 for CPdx's surveys 3-5 (September 2014 to August 2015) and 0 for survey waves 1 and 2 (January to August 2014). Column 1 includes household's expenditure on clothing and footwear, cosmetics, accessories, and beauty goods and services; column 2 includes items complementary to domestic chores of cooking and cleaning such as food, utensils and toiletries; column 3 includes time saving appliances (kitchen appliances) and services (domestic help). Column 4 includes expenditure on tobacco products and liquor, shaving articles and fuel for purposes other than cooking. Column 5 analyzes total expenditure on education such as fees to schools, colleges/ private tuition, books, uniforms etc. The second row includes standard errors in parentheses that are clustered by household. The third row reports p values in square brackets and fourth row reports sharpened two-stage q-values that correct the p-value of the interaction coefficient for false discovery rate from testing multiple hypotheses. The last two rows are a test for joint significant of differences between treated and comparison groups before policy reporting the chi-squared statistic and corresponding p value. The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave. The estimation controls for time varying characteristics of households and includes household and survey wave fixed effects. CG means comparison group mean. $*$ p< 0.1, $**$ p< .05, $***$ p< .01 .

Table 5: Aggregate treatment effect: Wife's account ownership and miscellaneous consumption

	Appliances	Restaurants	Entertainment	Rent & Utilities	Transport	Communication	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: per capita monthly expenditure in real terms							
Post ×	-1.997***	-0.785	0.322**	1.718***	-12.778***	-2.822***	2.083***
Treatment	(0.325)	(1.034)	(0.156)	(0.331)	(1.073)	(0.783)	(0.695)
	[0]	[0.448]	[0.039]	[0]	[0]	[0]	[0.003]
Sharpened q-value	0.001	0.069	0.014	0.001	0.001	0.001	0.002
CG mean	1.79	32.88	0.99	3.58	35.43	64.56	18.97
Joint significance of differences (pre-policy)							
Chi-squared value	5.721	0.322	0.461	0.003	25.496	4.564	2.391
p-value	0.017	0.57	0.497	0.958	0	0.033	0.122
Panel B: Share of total monthly expenditure							
Post ×	-0.001***	-0.001***	0.0001	-0.006***	0.0004	0.002***	0.001***
Treatment	(0.0001)	(0.0005)	(0)	(0.001)	(0.0004)	(0.0003)	(0.0001)
	[0]	[0.034]	[0.121]	[0]	[0.363]	[0]	[0]
Sharpened q-value	0.001	0.021	0.05	0.001	0.116	0.001	0.001
CG mean	0.001	0.022	0.	0.023	0.041	0.013	0.002
Joint significance of differences (pre-policy)							
Chi-squared value	3.234	0.182	4.316	25.747	10.036	3.408	0.016
p-value	0.072	0.67	0.038	0	0.002	0.065	0.899
Observations	48165	48165	48165	48165	48165	48165	48165
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the effect of wife's account ownership on miscellaneous consumption using an augmented inverse probability weighted DiD. Dependent variable is per capital monthly consumption in panel A and share of total consumption in panel B and specified in column titles. The first row reports aggregate treatment effect. The treated group includes households where female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation. The post-treatment dummy variable is assigned 1 for CPdx's surveys 3-5 (September 2014 to August 2015) and 0 for survey waves 1 and 2 (January to August 2014). Column 1 includes kitchen, household and mobile appliances; column 2 includes expenditure on food and drinks in restaurants, canteens and food stalls; column 3 includes tickets for movies, theatre, music concerts, sport events and other amusement activities; column 4 includes house rent and utilities; column 5 consists of modes of transport; column 6 includes phones, TV, internet and other media. Column 7 includes expenditure on medicines, doctor's/physiotherapists and hospitalization fees, medical tests and health insurance. The second row reports standard errors in parentheses that are clustered at the household level. The third row reports p values in square brackets and fourth row reports sharpened two- stage q-values that correct the p-value of the interaction coefficient for the false discovery rate from testing multiple hypotheses. The last two rows are a test for joint significant of differences between treated and comparison groups before policy. They report the chi-squared statistic and corresponding p value. The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave. The estimation controls for time varying characteristics of households and includes household and survey wave fixed effects. CG means comparison group mean. $*$ p< 0.1, $**$ p< .05, $***$ p< .01 .

Table 6: DiD: Effect of policy on expansion of account ownership between High and Low Impact districts

Dependent variable: Percent difference in accounts since 2014		
	(1)	(2)
High impact × Post	9.686***	10.456***
	(2.761)	(2.112)
Observations	3156	3132
R^2	0.921	0.951
Control group mean	-18.381	-18.381
District FE	Yes	Yes
Year FE	Yes	Yes
State*Year FE	No	Yes
LASSO controls	Yes	Yes

Notes: The table reports the average expansion of bank accounts since 2014 between High and Low Impact district after policy. High impact includes districts at state median bank branch density and below. Low impact includes the districts above state median. The estimation includes district, year and state*year fixed effects and controls selected by post double (LASSO) selection method (Belloni et al., 2013). The control variables are number of primary schools and senior secondary schools in towns in 2011, changes in the rural population, composition of scheduled tribes and length of paved roads connecting villages as well as the change in hours of electricity supply to domestic users. Standard errors are reported in parentheses and clustered by district. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 7: Parallel trends test: Effect of expansion in account ownership on women's empowerment

	Alone/ joint decision on big household purchases		Money available for autonomous use	
	(1)	(2)	(3)	(4)
High impact × Pre-treatment time dummy	-0.010 (0.018)	-0.008 (0.014)	-0.005 (0.012)	-0.006 (0.011)
Observations	49005	50132	50230	50230
R^2	0.188	0.219	0.074	0.130
Control group mean	0.508	0.508	0.863	0.863
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
LASSO controls included	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes

Notes: This table reports differences in empowerment outcomes between High and Low Impact districts in the years before policy implementation. Dependent variables are listed as column titles. The decision making variables - whether woman participates jointly or with spouse on decisions to purchase big household items, and has money available for autonomous use - are binary. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Only coefficients of interaction terms of High Impact districts with year dummy are reported. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS 2005 and 2012), bank branch density is estimated using data from RBI and Population Census. All specifications include district fixed effects and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 8: DiD: Effect of expansion in account ownership on women's empowerment

	Alone/ joint decision on big household purchases		Money available for autonomous use	
	(1)	(2)	(3)	(4)
High impact × Post	0.006	-0.001	-0.039*	-0.038**
	(0.021)	(0.017)	(0.023)	(0.017)
Observations	35463	35463	36727	36727
R^2	0.252	0.259	0.345	0.356
Control group mean	0.650	0.650	0.615	0.615
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
LASSO controls included	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes

Notes: This table reports differences in empowerment outcomes between High and Low Impact districts before and after policy implementation. Dependent variables are listed as column titles. The decision making variables - whether woman participates jointly or with spouse on decisions to purchase big household items, and has money available for autonomous use - are binary. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Post is a binary variable estimating differences before and after policy. Post is 1 for observations from DHS 2015-16 and 0 for IHDS 2011-12. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS, DHS), bank branch density is estimated using data from RBI and Population Census. Districts not surveyed in both IHDS and DHS are excluded from analysis. All specifications include district and year fixed effects and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Columns 2 and 4 also control for state × year fixed effects. Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 9: DiD: Other empowerment outcomes

	Employed		Fertility		Mobility outside home		Cruelty by spouse/ his family	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High impact × Post-treatment time dummy	-0.047**	-0.045**	-0.034	-0.034	-0.019	-0.021	-12.392	-10.896
	(0.020)	(0.017)	(0.023)	(0.022)	(0.021)	(0.017)	(8.932)	(7.956)
Observations	67881	67881	67148	67148	36370	36370	1821	1815
R^2	0.227	0.235	0.493	0.494	0.115	0.128	0.931	0.942
Control group mean	0.384	0.384	2.221	2.221	0.091	0.091	190.447	190.447
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes	No	Yes	No	Yes

Notes: This table reports differences in empowerment outcomes between High and Low Impact districts before and after policy implementation. Dependent variables are listed as column titles - respondent is employed, women's fertility (number of children), ability to visit friends/relatives and public spaces on her own and annual dowry-related violence cases in a district. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Post is a binary variable estimating differences before and after policy. Post is 1 for observations from DHS 2015-16 and 0 for IHDS 2011-12. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS, DHS), bank branch density is estimated using data from RBI and Population Census and crimes data is extracted from the National Crime Records Bureau. Districts not surveyed in both IHDS and DHS are excluded from analysis. All specifications include district and year fixed effects and columns 1-6 include controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Columns 2, 4, 6 and 8 also control for state × year fixed effects. Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Appendix A Data and variables

Table A.1: Description of raw indicators: household resource allocation

Variable	Items	Description
Women's consumption	Clothing and Footwear	Garments, Footwear and accessories such as accessories such like socks, tie, scarves, handkerchiefs, fabric and tailoring
	Accessories	Artificial jewelry, bags, wallets, watches, goggles, glasses and gems and jewelry
	Cosmetics	Includes 'dental care products', 'bathing soap', 'face wash', 'shaving articles', 'hair oil', 'shampoo and hair conditioner', 'powder', 'creams', 'deodorants and perfumes', 'henna, hair color, hair gel, etc.', 'lipstick and other cosmetics'
	Beauty products and services	Beauty enhancement services like beauty parlors, hair stylists, barber services, salons and spas, masseur services, etc.
Cooking & Sanitation	Food	Includes cereals & pulses, edible oils, spices, vegetables & fruits, meat, fish & eggs, milk & milk products, ready-to-eat food, spices, bread, biscuits, namkeens & salty snacks, noodles & pasta, flakes, muesli & oats, confectionery & ice-creams, health supplements, tea, coffee, sweeteners, and beverages, juices & bottled water
	Utensils	Cookware including cups, saucers, plates, spoons, containers, frying pans etc. and kitchen accessories including kitchen knives, cutting plates, sieves (chalni), can openers, coffee filters, etc.
	Toiletries	Includes detergent bar,powder,liquids, scourer and housecleaning agents, and other housecare products
Time saving	Time saving appliances and services	Gadgets such as toasters, water filters, microwave oven, refrigerator, cooking range, stove, mixer/-grinder, juicer, coffee machine, grill, induction, chimney, exhaust system and any other appliances that are used in kitchens to improve the efficiency of cooking
		Salary paid to maid servants, cooks, drivers, guards, gardeners, baby sitters and other staff, and laundry services

Men's consumption	Shaving articles	Shaving brush, razors, blades, shaving foams, shaving gels, shaving lather sticks, after shave lotions and shaving creams
	Intoxicant	Cigarettes, 'bidis', other tobacco products and liquor
	Fuel other than cooking	Petrol and diesel or any other petroleum fuel product for its own consumption
Education	School and college fees	Admission and exam fees, uniform, lab and library fees, extra classes, use of sport facilities etc.
	Private coaching	Private tutors or classes
	Stationery	Notebooks, writing pads, paper, pens and pencils, markers, erasers, rulers, compass set, pins, staplers, post-it and related articles
Formal Savings	Formal sources	- Bank fixed deposits - Schemes offered by India Post (eg. Post Office Savings Account, Post Office Time Deposit Account, Senior Citizen Savings Scheme etc.) - Government bonds and Public Provident Funds - Kisan Vikas Patra - Employee Provident Fund
Saving SHG/ MFI	Microfinance	- Self-Help Groups - Chit funds - Microfinance Institution
Formal Borrowing	Household has outstanding debt in formal sources	- Banks - Employer - Registered companies engaged in the business of loans and advances, insurance business or chit business - Microfinance Institution - Credit card
Informal Borrowing	Informal sources	- Moneylender - Relatives or friends - Shops - Other sources such as non-professional moneylender, religious institutions and missionaries etc.
Gold	Gold	- Gold assets or funds including gold bars, ornaments, jewelry and Gold Exchange Traded Funds
Investment	Household has outstanding investment	- Mutual funds - Listed shares - Private business enterprise including equity capital of an unlisted company, limited liability partnership or a contribution to a partnership or a proprietorship concern - Real investment including house, plot of land, apartment, bungalow, office space, shop or farmhouse

The variables used to measure self-reports of women's decision making are - participation in household's big purchase decisions, autonomous use of money and violence from spouse or his relatives. These variables are extracted from IHDS and DHS. Table A.2 lists the survey questions used to create the indicators. The first indicator, women's self reported participation in household's decisions on purchase of big/ expensive items is constructed as binary indicator which assigns the value 1 when women respond if they make the decision independently or jointly with their spouse and 0 if the spouse and/or other members of the household make this decision. The indicator constructed for analysis is essentially a measure of joint decision making because the DHS asks respondents if she or other members "usually" make this decision, failing to capture whether the woman makes this decision independently always. The second indicator asks respondents if they have access to any money that they can autonomously decide how to spend. The variables in IHDS and DHS allow for a lower bound estimate of women's autonomous use of money as they differ in the source of money and type of expenditure they capture. The DHS asks if the money available to the woman is her own while the responses captured in IHDS could potentially include cash from another member's earnings. The IHDS specifically asks about household expenditures while the DHS does not limit the use of money by type of expenditure. The binary indicator constructed from these two variables is assigned 1 when they respond yes to a question and 0 if they respond no. Both surveys, IHDS and DHS, interview married women between the ages 15 and 49.

Table A.2: Description of raw indicators: women's self-reported participation in decision-making

	Participation in purchase decisions	Autonomous use of money
	(1)	(2)
IHDS (2005, 2012)	Who in your family decides the following: Whether to buy an expensive item such as a TV or fridge?	Do you yourself have any cash in hand to spend on household expenditures?
DHS (2015-16)	Who usually makes decisions about making major household purchases: mainly you, mainly your husband, you and your husband jointly, or someone else?	Do you have any money of your own that you alone can decide how to use?

The following variables measured using the IHDS 2012 round are included in the analysis of mechanisms through which account ownership can affect women's empowerment. All indicators of type "Count" are generated by counting the number of "yes" for the constituent variables.

Table A.3: Description of mechanisms

Indicator	Type	Raw variables
Spousal cooperation	Count	1. Respondent and spouse discuss things that happen at work/farm 2. Respondent and spouse discuss what to spend money on 3. Respondent and spouse discuss things about the community such as elections or politics
Wife beating	Count	1. Is it usual for husband to beat wife if she goes out without telling him? 2. Is it usual for husband to beat wife if her natal family does not give gifts (money/ jewellery)? 3. Is it usual for husband to beat wife if she neglects house or child? 4. Is it usual for husband to beat wife if she doesn't cook food properly? 5. Is it usual for husband to beat wife if he suspects her of having relations with other men?
Mobility	Count	1. Health centre 2. Friends and relatives 3. Local convenience store
Education	Continuous	Years of completed schooling
Employment	Binary	Whether respondent is employed
TrustBanks	Count	1. Whether household took loan from bank in last 5 years 2. Whether household has confidence in bank to keep money safe
Transfers	Binary	Whether household received government transfers

Appendix B Figures

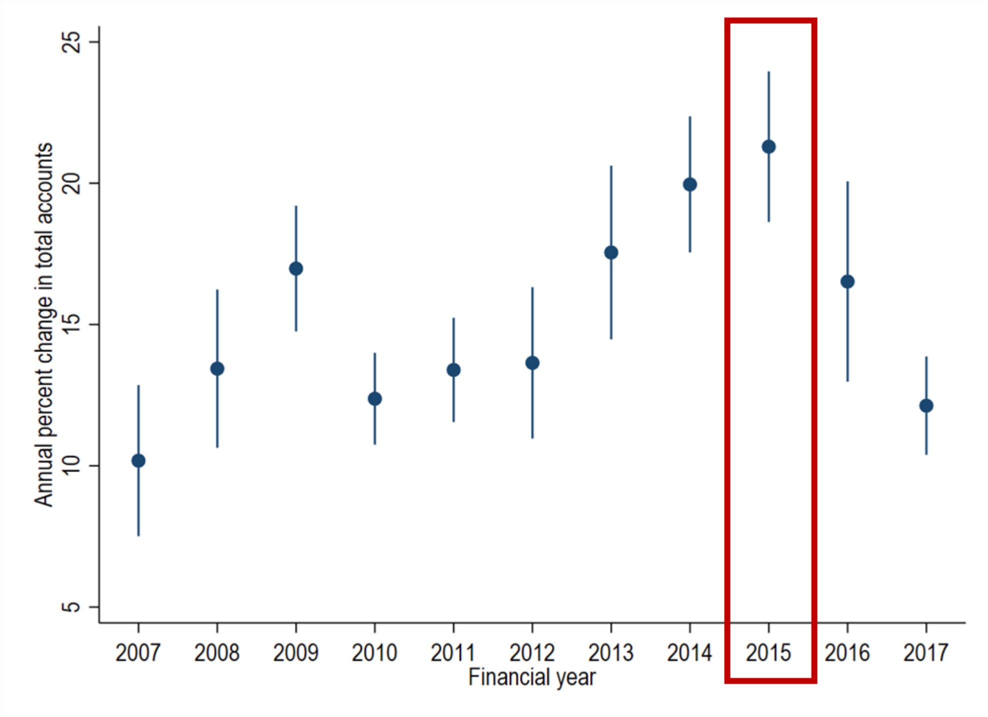

Figure B.1: Percent annual increases in account ownership

Notes: This figure shows that the annual percentage increase in bank accounts for a given district was largest after PMJDY policy. The blue dots depict the average percentage change in total bank accounts for a district by financial year and the vertical lines represent 95% confidence intervals. The financial year 2015 starts April 1, 2014 and ends March 31, 2015. Source: Reserve Bank of India. Author's calculations.

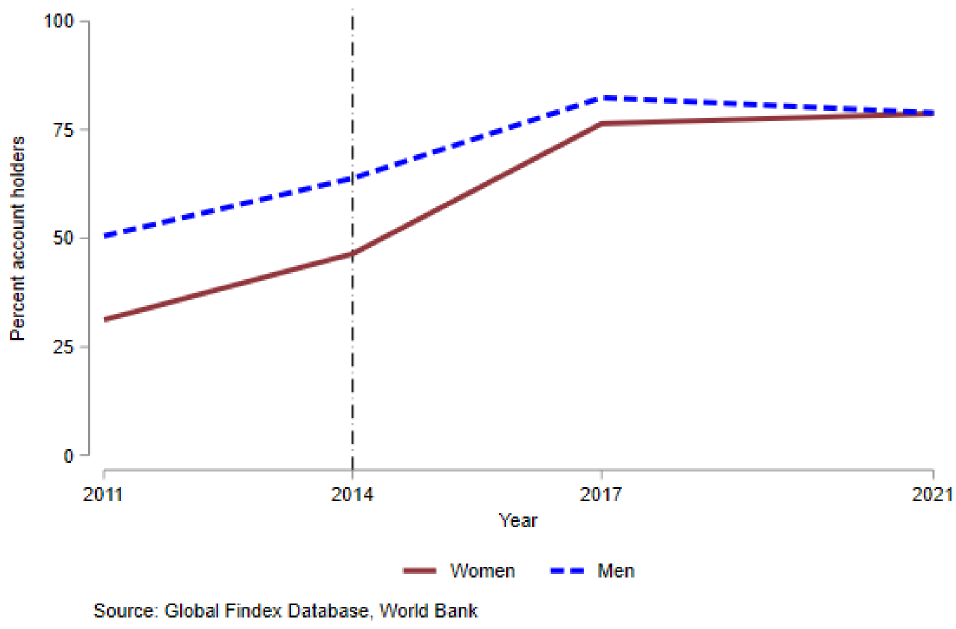

Figure B.2: Trends in bank account ownership by sex

Notes: This figure reports the percentage share of bank accounts by sex over time. The vertical dash line signifies year PMJDY was implemented. The gender gap in account ownership started decreasing after 2011, and further reduced after the 2014 policy. Source: World Bank Findex Data for India. Author's calculations.

Appendix C Tables

Table C.1: Differences in household borrowing, saving and investment between treatment and comparison groups by survey-wave

	Formal Saving (1)	Saving SHG/MFI (2)	Formal Borrowing (3)	Informal Borrowing (4)	Gold Savings (5)	Inves-tment (6)
Jan-Apr 2014	-0.004	-0.005*	-0.002	0.000	0.000	-0.013*
	(0.006)	(0.003)	(0.002)	(0.004)	(0.006)	(0.007)
Sep-Dec 2014	0.100***	0.006**	0.008***	-0.001	0.012*	-0.010
	(0.009)	(0.003)	(0.003)	(0.004)	(0.006)	(0.007)
Jan-Apr 2015	0.110***	0.007***	0.004	-0.016***	0.014**	0.003
	(0.009)	(0.003)	(0.003)	(0.004)	(0.006)	(0.007)
May-Aug 2015	0.134***	0.010***	0.002	-0.003	0.023***	0.013*
	(0.010)	(0.003)	(0.003)	(0.004)	(0.007)	(0.008)
Observations	64559	64559	64559	64559	64559	64559
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimated differences in household's uptake of financial services between the treated and comparison groups by survey wave from the augmented inverse-probability weighted DiD. The first row provides evidence of parallel trends before policy. The survey wave, from May through August is omitted as the reference group. The dependent variables are specified in column titles and are binary indicators of whether- household has saved in bank/ post office deposits or government bonds (column 1), savings in Self-Help group, Chit fund, microfinance institute (column 2), outstanding borrowing from a bank/ registered company (column 3), loans from friends, moneylender and other informal sources (Column 4), saved in gold/ related assets (column 5) and invested in shares, mutual funds/ real estate (column 6). The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave. The treated group includes households where the female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation (waves 1-5). The outcome and treatment models include covariates listed in Table 1. The estimation includes household fixed effects. Standard errors are clustered by household and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table C.2: Differences in consumption allocations between treatment and comparison groups by survey wave

	Women's consumption	Cooking & sanitation	Time saving	Men's consumption	Education
	(1)	(2)	(3)	(4)	(5)
Panel A: Per capita monthly consumption (real terms)					
Jan-Apr14	-0.426	-2.435	-0.804***	-10.668***	0.629
	(4.420)	(6.262)	(0.236)	(2.899)	(1.909)
Sep-Dec14	11.547**	11.943*	-1.042***	-14.410***	-8.273***
	(4.684)	(6.425)	(0.190)	(3.232)	(2.225)
Jan-Apr15	-2.866	-24.615***	-0.773***	-16.226***	-11.861***
	(4.616)	(6.674)	(0.198)	(3.460)	(2.424)
May-Aug15	-3.561	-26.470***	-1.543***	-4.334	-13.813***
	(4.834)	(7.136)	(0.229)	(3.719)	(2.468)
Panel B: Share of total monthly consumption					
Jan-Apr14	0.000	0.006*	-0.000***	-0.001	0.001
	(0.002)	(0.003)	(0.000)	(0.001)	(0.001)
Sep-Dec14	0.006***	0.011***	-0.000***	-0.004***	-0.005***
	(0.002)	(0.003)	(0.000)	(0.001)	(0.001)
Jan-Apr15	0.001	0.021***	-0.000***	-0.003**	-0.006***
	(0.002)	(0.003)	(0.000)	(0.002)	(0.001)
May-Aug15	0.001	0.014***	-0.001***	0.002	-0.007***
	(0.002)	(0.003)	(0.000)	(0.002)	(0.001)
Observations	48165	48165	48165	48165	48165
Covariates	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimated differences in per capita monthly consumption allocations (panel A) and share of total monthly consumption (panel B) between the treated and comparison groups by survey wave from the augmented inverse-probability weighted DiD. The first row provides evidence of parallel trends before policy. Survey wave May through August is omitted as the reference group. The dependent variables are household's per capita monthly consumption expenses in real terms and are listed as column titles. Column 1 includes household's expenditure on clothing and footwear, cosmetics, accessories, and beauty goods and services; column 2 includes items complementary to domestic chores of cooking and cleaning such as food, utensils and toiletries; column 3 includes time saving appliances (kitchen appliances) and services (domestic help). Column 4 includes expenditure on tobacco products and liquor, shaving articles and fuel for purposes other than cooking. Column 5 analyzes total expenditure on education such as fees to schools, colleges/ private tuition, books, uniforms etc. The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave. The treated group includes households where the female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation (waves 1-5). The outcome and treatment models include covariates listed in Table 1. The estimation includes household fixed effects. Standard errors are clustered by household and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table C.3: DiD: Varying definition of High/Low Impact districts

	Discrete classification		Continuous
	Government owned	Private owned	Branch density
	(1)	(2)	(3)
Panel A: Percent difference in accounts since 2014			
High impact × Post	4.080*	8.313***	-1.198***
	(2.319)	(1.935)	(0.212)
Observations	2871	3553	3597
R^2	0.950	0.953	0.954
Control group mean	-18.344	-18.160	.
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
LASSO controls	Yes	Yes	Yes
Panel B: Alone/ joint decision on big household purchases			
High impact × Post	0.008	-0.020	0.001
	(0.017)	(0.017)	(0.001)
Observations	35451	35399	35399
Control group mean	0.656	0.650	.
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
LASSO controls	Yes	Yes	Yes
Panel C: Money available for autonomous use			
High impact × Post	-0.021	-0.030*	0.005***
	(0.017)	(0.017)	(0.002)
Observations	36714	36714	37615
Control group mean	0.606	0.615	.
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
LASSO controls	Yes	Yes	Yes

Notes: This table reports aggregate differences between High and Low Impact districts using Eq. 4 for varying definitions of High/Low Impact. Panel A tests the effects on growth in bank accounts since 2014 and panels B and C on women's participation in decision making. Columns 1 and 2 define High/Low Impact as less/greater than median branch density of government and private owned banks, respectively. Column 3 tests effects of continuous variable of branch density. Post is a time dummy estimating differences before and after policy. All estimations include district, time and state × year fixed effects, and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). There are only two time periods in Panels B and C. Standard errors are clustered by district and reported in parentheses. There are no districts with 0 bank branches, therefore, control group mean is missing in column 3. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table C.4: DiD: Effect of account expansion policy on account ownership in survey data

	Proportion households banked		Proportion women banked	
	(1)	(2)	(3)	(4)
	Panel A: Pre-treatment changes			
High impact ×	-0.003	-0.014	-0.039**	-0.043***
Pre-treatment time dummy	(0.019)	(0.016)	(0.017)	(0.014)
Observations	40522	37034	36241	36286
R^2	0.563	0.719	0.476	0.656
Control group mean	0.552	0.552	0.291	0.291
	Panel B: Post-policy			
High impact × Post	-0.003	0.007	0.031*	0.041**
	(0.020)	(0.019)	(0.019)	(0.016)
Observations	33718	33718	35607	35607
R^2	0.853	0.888	0.880	0.913
Control group mean	0.909	0.909	0.119	0.119
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
LASSO controls	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes

Notes: This table reports differences in proportion of banked households (columns 1-2) and women (columns 3-4) between High and Low Impact districts. Panel A estimates differences before policy using IHDS 2005 and 2012, while Panel B estimates impact of policy using IHDS 2012 and DHS 2015-16. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Post is a binary variable that assigns 1 to observations from household survey after policy implementation (DHS 2015-16) and 0 to observations from survey before policy implementation (IHDS 2012). Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS, DHS), and bank infrastructure is estimated using data from RBI and Population Census. Districts not surveyed in both IHDS and DHS are excluded from analysis. All specifications include district fixed effects and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Columns 2 and 4 also include state × year fixed effects. Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table C.5: DiD: Account expansion and women's empowerment by quartiles of bank branch density

	Alone/ joint decision on big household purchases	Money available for autonomous use
	(1)	(2)
Quartile 2 × Post	-0.008	-0.025
	(0.024)	(0.023)
Quartile 3 × Post	0.003	0.014
	(0.022)	(0.025)
Quartile 4 × Post	0.017	0.053**
	(0.021)	(0.024)
Observations	36294	37615
R^2	0.276	0.366
Control group mean	0.650	0.615
District FE	Yes	Yes
Year FE	Yes	Yes
State × Year FE	Yes	Yes
LASSO controls	Yes	Yes

Notes: This table reports aggregate differences between districts by quartiles of ex-ante bank branch density. The first quartile is used as reference category. Dependent variables are listed as column titles and are binary. Post is a time dummy estimating differences before and after policy. All estimations include district, time and state × year fixed effects, and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table C.6: Heterogeneous treatment effects: Cooperation between spouses

	Spousal cooperation				Acceptability of wife beating			
	Alone/ joint decision on big household purchases		Money available for autonomous use		Alone/ joint decision on big household purchases		Money available for autonomous use	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Characteristic × High Impact × Post	-0.009	-0.030	-0.005	-0.018	-0.011	-0.005	-0.011	0.005
	(0.057)	(0.043)	(0.059)	(0.045)	(0.023)	(0.017)	(0.022)	(0.020)
High impact × Post	0.037	0.094	-0.027	0.013	0.032	0.012	-0.015	-0.050
	(0.177)	(0.134)	(0.183)	(0.140)	(0.058)	(0.046)	(0.053)	(0.052)
Observations	35463	35463	36727	36727	35463	35463	36727	36727
R^2	0.252	0.259	0.345	0.356	0.253	0.259	0.347	0.356
Control group mean	0.650	0.650	0.615	0.615	0.650	0.650	0.615	0.615
MDE (80% power)	0.159	0.120	0.165	0.126	0.064	0.049	0.062	0.055
Standardized effect	-0.019	-0.063	-0.010	-0.037	-0.023	-0.010	-0.023	0.010
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes	No	Yes	No	Yes
LASSO controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table tests for heterogeneity in women's decision making between High and Low Impact districts in response to the policy. Dependent variables - whether woman participates jointly or with spouse on decisions to purchase big household items, and has money available for autonomous use - are binary. "Characteristic" specifies the heterogeneity variable which are district-wise ex-ante aggregates of cooperation between spouses (results in columns 1-4) and acceptability of wife beating (columns 5-8). Description of these scores is provided in Appendix Table A.3. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Post is a binary variable estimating differences before and after policy. Post is 1 for observations from DHS 2015-16 and 0 for IHDS 2011-12. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS, DHS), and bank branch density is estimated using data from RBI and Population Census. Districts not surveyed in both IHDS and DHS are excluded from analysis. All specifications include district and year fixed effects, and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Columns 2, 4, 6 and 8 also include state× year fixed effects. Standard errors are clustered by district and reported in parentheses. ∗ p< 0.1, ∗∗ p< .05, ∗ ∗ ∗ p< .01

Table C.7: Heterogeneous treatment effects: Account operability

	Mobility outside home				Woman's education			
	Alone/ joint decision on big household purchases		Money available for autonomous use		Alone/ joint decision on big household purchases		Money available for autonomous use	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Characteristic × High Impact × Post	0.075*	0.064	-0.018	-0.015	0.004	0.007	-0.022**	-0.022**
	(0.045)	(0.042)	(0.049)	(0.044)	(0.010)	(0.008)	(0.010)	(0.009)
High impact × Post	-0.160	-0.145	0.004	-0.006	-0.009	-0.043	0.102*	0.087*
	(0.099)	(0.093)	(0.110)	(0.096)	(0.057)	(0.047)	(0.062)	(0.051)
Observations	35463	35463	36727	36727	35463	35463	36727	36727
R^2	0.252	0.260	0.346	0.356	0.252	0.259	0.346	0.356
Control group mean	0.650	0.650	0.615	0.615	0.650	0.650	0.615	0.615
MDE (80% power)	0.125	0.116	0.138	0.122	0.027	0.023	0.029	0.025
Standardized effect	0.157	0.134	-0.037	-0.031	0.008	0.015	-0.045	-0.045
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes	No	Yes	No	Yes
LASSO controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table tests for heterogeneity in women's decision making between High and Low Impact districts in response to the policy. Dependent variables - whether woman participates jointly or with spouse on decisions to purchase big household items, and has money available for autonomous use - are binary. "Characteristic" specifies the heterogeneity variable which are district-wise ex-ante aggregates of whether the respondent can visit friends/relatives and public spaces by herself, "mobility", (results in columns 1-4), and her years of completed schooling, "education", (columns 5-8). Description of these characteristics is provided in Appendix Table A.3. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Post is a binary variable estimating differences before and after policy. Post is 1 for observations from DHS 2015-16 and 0 for IHDS 2011-12. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS, DHS), bank branch density is estimated using data from RBI and Population Census. Districts not surveyed in both IHDS and DHS are excluded from analysis. All specifications include district and year fixed effects, and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Columns 2, 4, 6 and 8 also include state× year fixed effects. Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table C.8: Heterogeneous treatment effects: Account use

	Respondent is employed				Trust in banking				Household receives transfers			
	Alone/ joint decision on big household purchases		Money available for autonomous use		Alone/ joint decision on big household purchases		Money available for autonomous use		Alone/ joint decision on big household purchases		Money available for autonomous use	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Characteristic × High Impact × Post	-0.053	-0.052	0.080	0.101	0.206	0.180	-0.129	-0.214	0.036	0.052	0.081	0.078
	(0.100)	(0.081)	(0.100)	(0.083)	(0.142)	(0.127)	(0.170)	(0.132)	(0.143)	(0.111)	(0.145)	(0.124)
High impact × Post	0.034	0.026	-0.083	-0.099*	-0.244	-0.221	0.118	0.222	-0.004	-0.016	-0.061	-0.061
	(0.063)	(0.051)	(0.062)	(0.050)	(0.171)	(0.156)	(0.205)	(0.158)	(0.049)	(0.037)	(0.046)	(0.037)
Observations	35463	35463	36727	36727	35463	35463	36727	36727	35463	35463	36727	36727
R^2	0.252	0.259	0.345	0.356	0.252	0.260	0.345	0.356	0.252	0.259	0.345	0.356
Control group mean	0.650	0.650	0.615	0.615	0.650	0.650	0.615	0.615	0.650	0.650	0.615	0.615
MDE (80% power)	0.279	0.227	0.280	0.232	0.397	0.356	0.475	0.371	0.400	0.312	0.405	0.348
Standardized effect	-0.111	-0.109	0.164	0.207	0.432	0.377	-0.265	-0.439	0.075	0.109	0.166	0.160
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
LASSO controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table tests for heterogeneity in women's decision making between High and Low Impact districts in response to the policy. Dependent variables - whether woman participates jointly or with spouse on decisions to purchase big household items, and has money available for autonomous use - are binary. "Characteristic" specifies the heterogeneity variable which are district-wise ex-ante aggregates of whether respondent is employed (results in columns 1-4), household's trust in banking institutions (columns 5-8) and whether household has received a government transfer (columns 9-12). Description of these characteristics is provided in Appendix Table A.3. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Post is a binary variable estimating differences before and after policy. Post is 1 for observations from DHS 2015-16 and 0 for IHDS 2011-12. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS, DHS), bank branch density is estimated using data from RBI and Population Census. Districts not surveyed in both IHDS and DHS are excluded from analysis. All specifications include district fixed effects and controls selected from Table 2 by post double (LASSO) selection method (Belloni et al., 2013). Columns 2, 4, 6, 8, 10 and 12 also include state× year fixed effects. Standard errors are clustered by district and reported in parentheses. * p< 0.1, ** p< .05, * * * p< .01

Online Appendix

The online appendix describes financial inclusion policies in India, sampling methodology and summary statistics of surveys used in this paper, and additional results.
Link to Appendix.

Disclosure statement

The author declares that she has no relevant or material financial interests that relate to the research described in this paper.